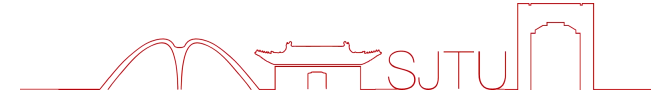




上海交通大学
SHANGHAI JIAO TONG UNIVERSITY



Federated Learning (FL) Research and Application

马汝辉
上海交通大学

饮水思源 · 爱国荣校



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Challenges in FL

2

Communication-efficiency

3

Privacy protection

4

Heterogeneity

5

Application



00

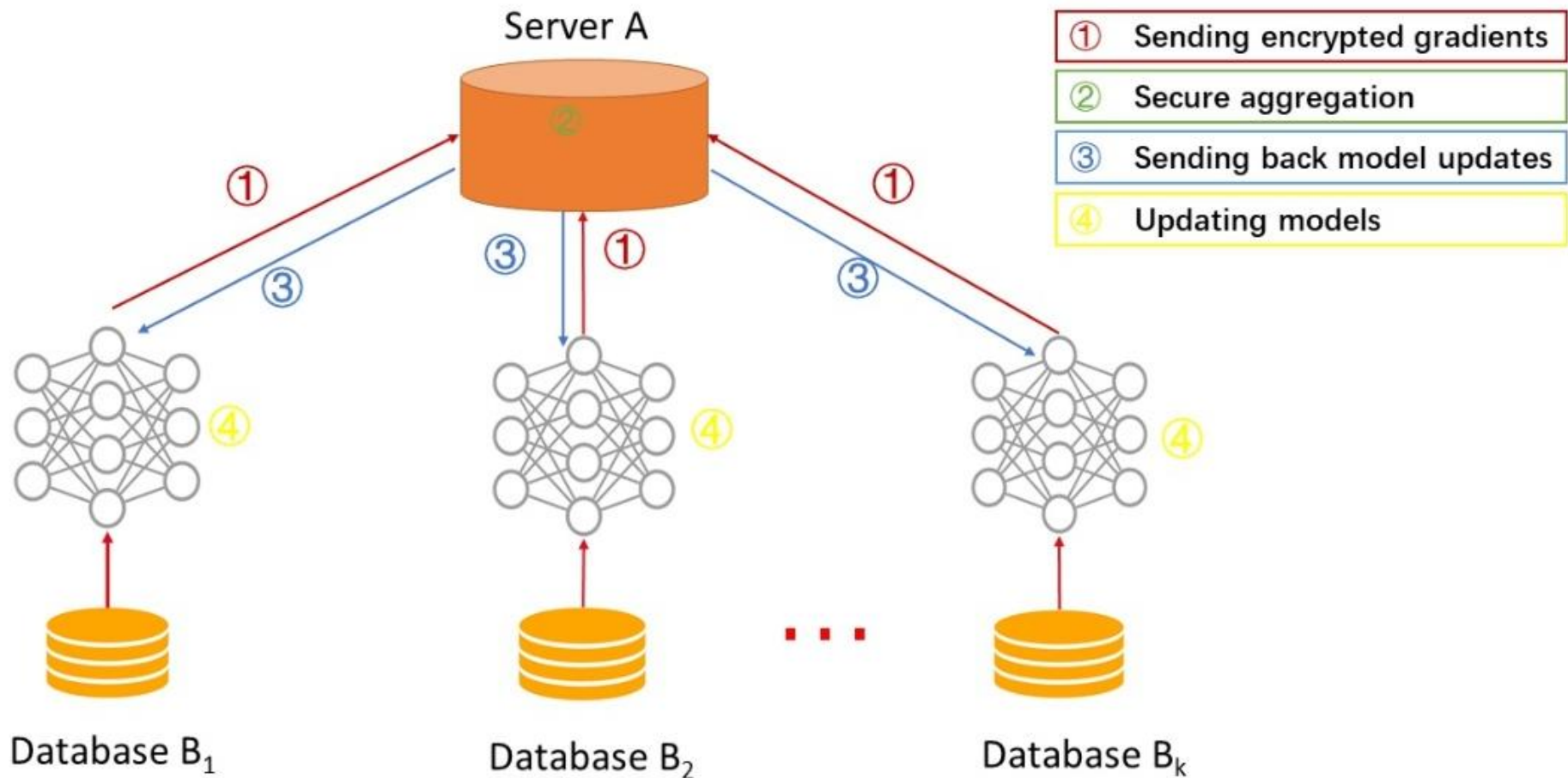
Preliminary

- Introduction
- Structure
- Procedure



联邦学习介绍

- ① 联邦学习是一种分布式机器学习技术，或机器学习框架。
- ② 在保证**数据隐私安全及合法合规**的基础上，实现**共同建模**，提升AI模型的效果。
- ③ 本质上是通过多个用户设备共同训练一个代表所有用户设备的**全局模型**
- ④ 训练过程不需要用户数据的交换，更强调隐私性。



总体架构

服务器端

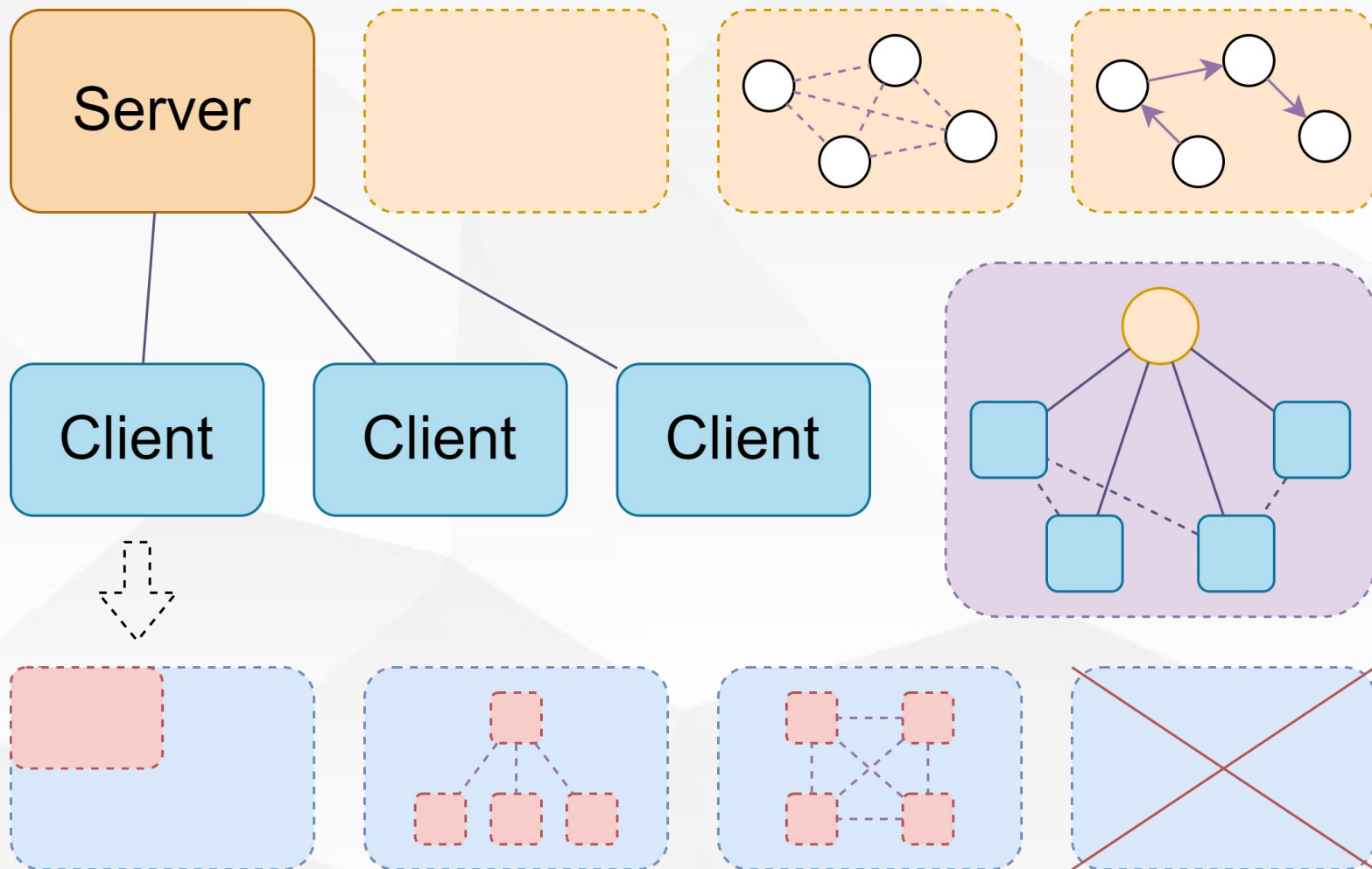
- 有、无 - 控制方式
- 单点、多点 - 一致性
- 区块链 - 可信

网络连接

- 去中心、中心、半中心

客户端

- 资源占用 - 全部、部分
- 拓扑结构 - 层次化
- 计算能力 - 有无



控制方式

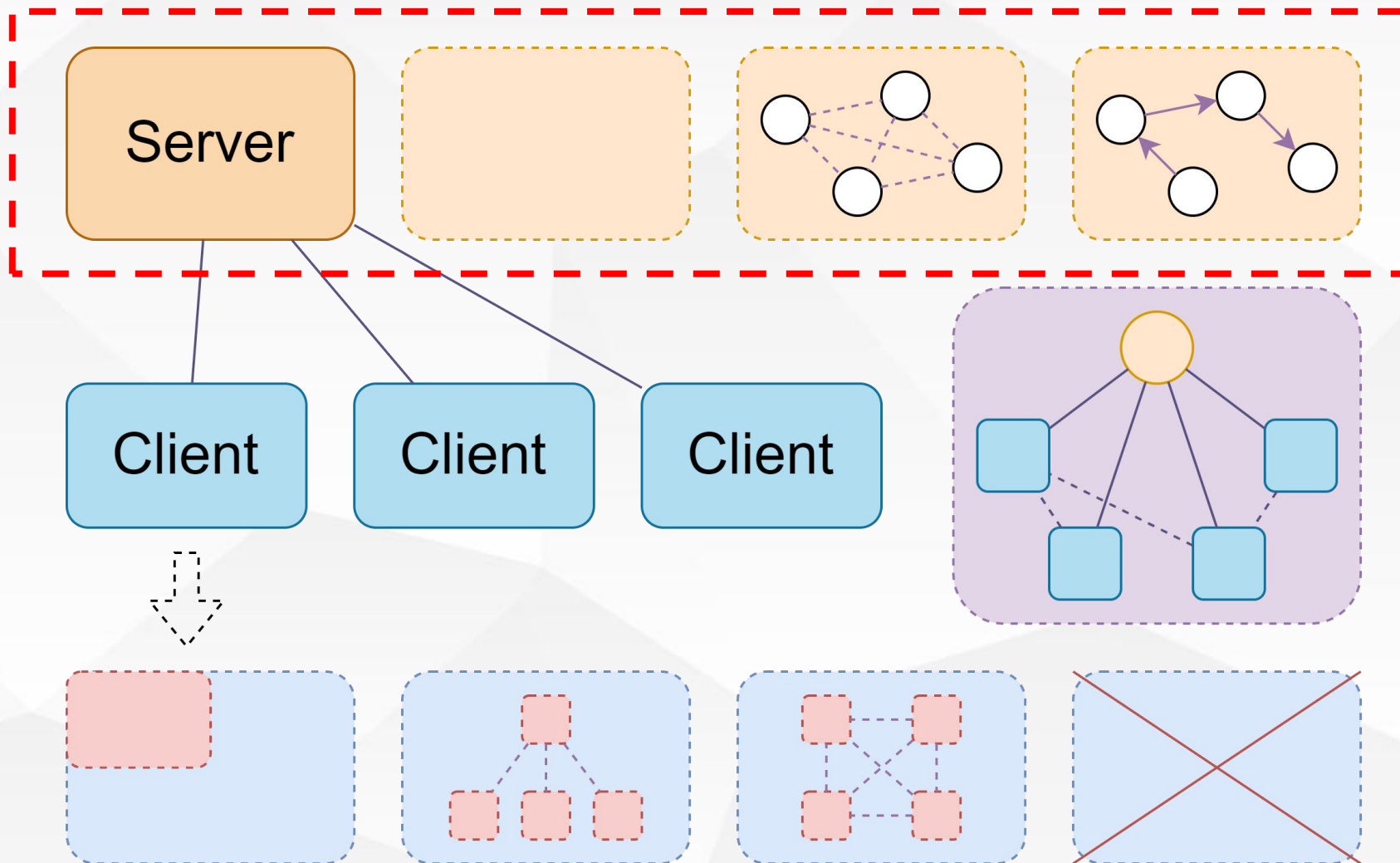
- 中心化服务器
- 去中心化控制

一致性

- 单点服务器
- 多点服务器集群

可信第三方

- 区块链服务器



① 中心化框架

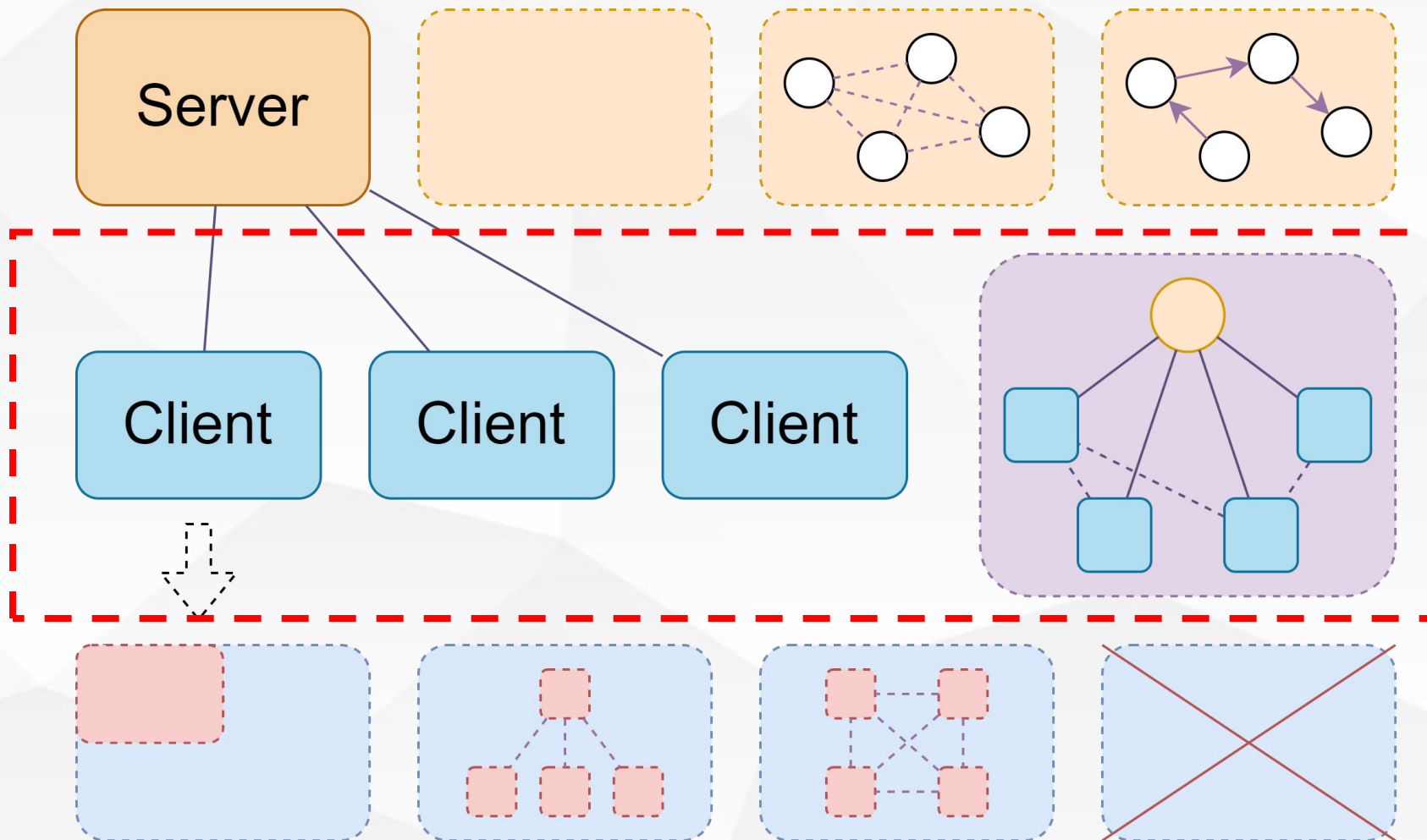
- 服务器端-客户端

② 去中心化框架

- 客户端-客户端

③ 半中心化框架

- 服务器端-客户端
- 客户端-客户端



资源占用

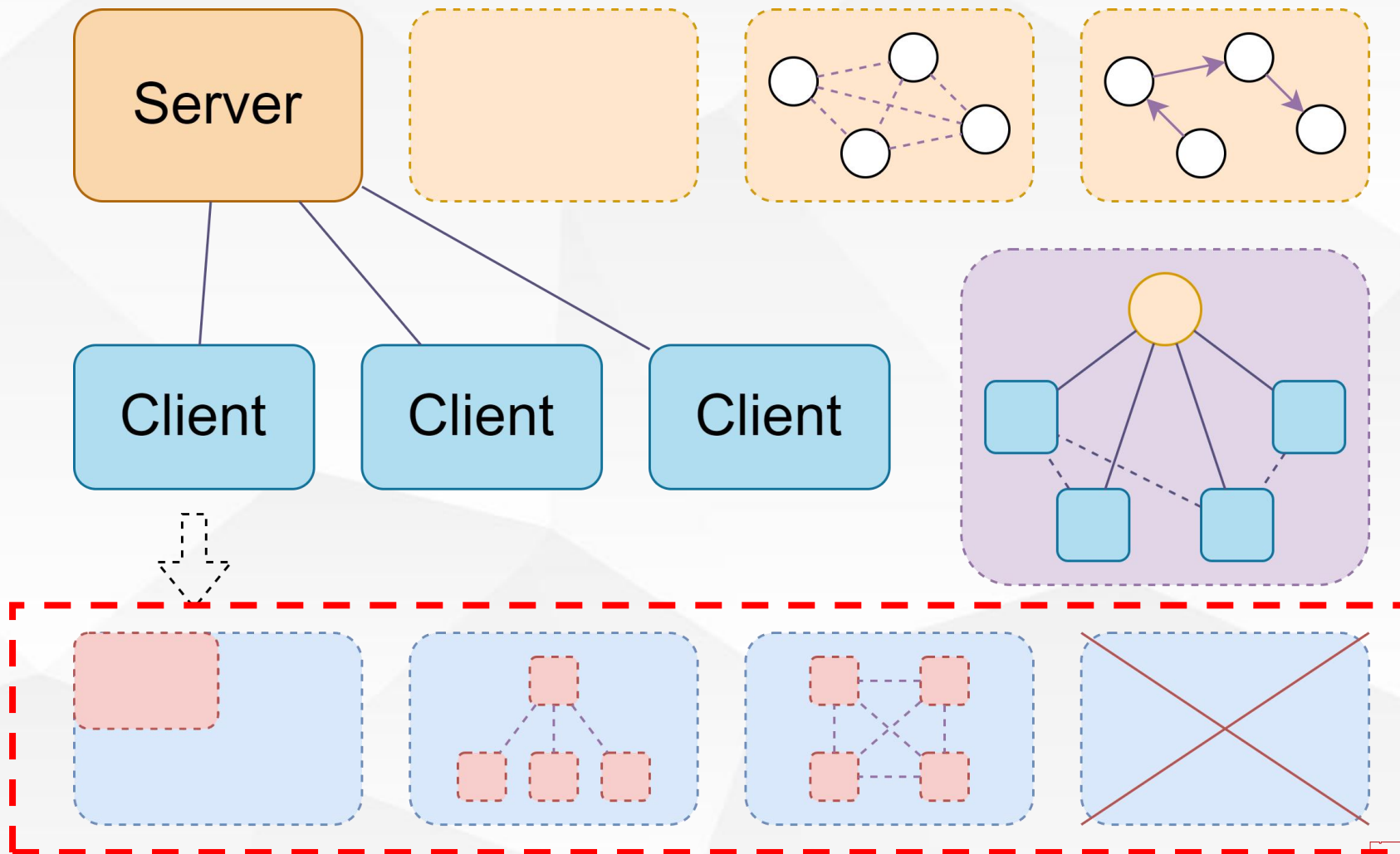
- 虚拟机/容器
- 资源隔离性

拓扑结构

- 多层中心化
- 集群分布式

设备能力

- 低计算速度
- 少存储空间





总体架构

任务层：上层应用角度

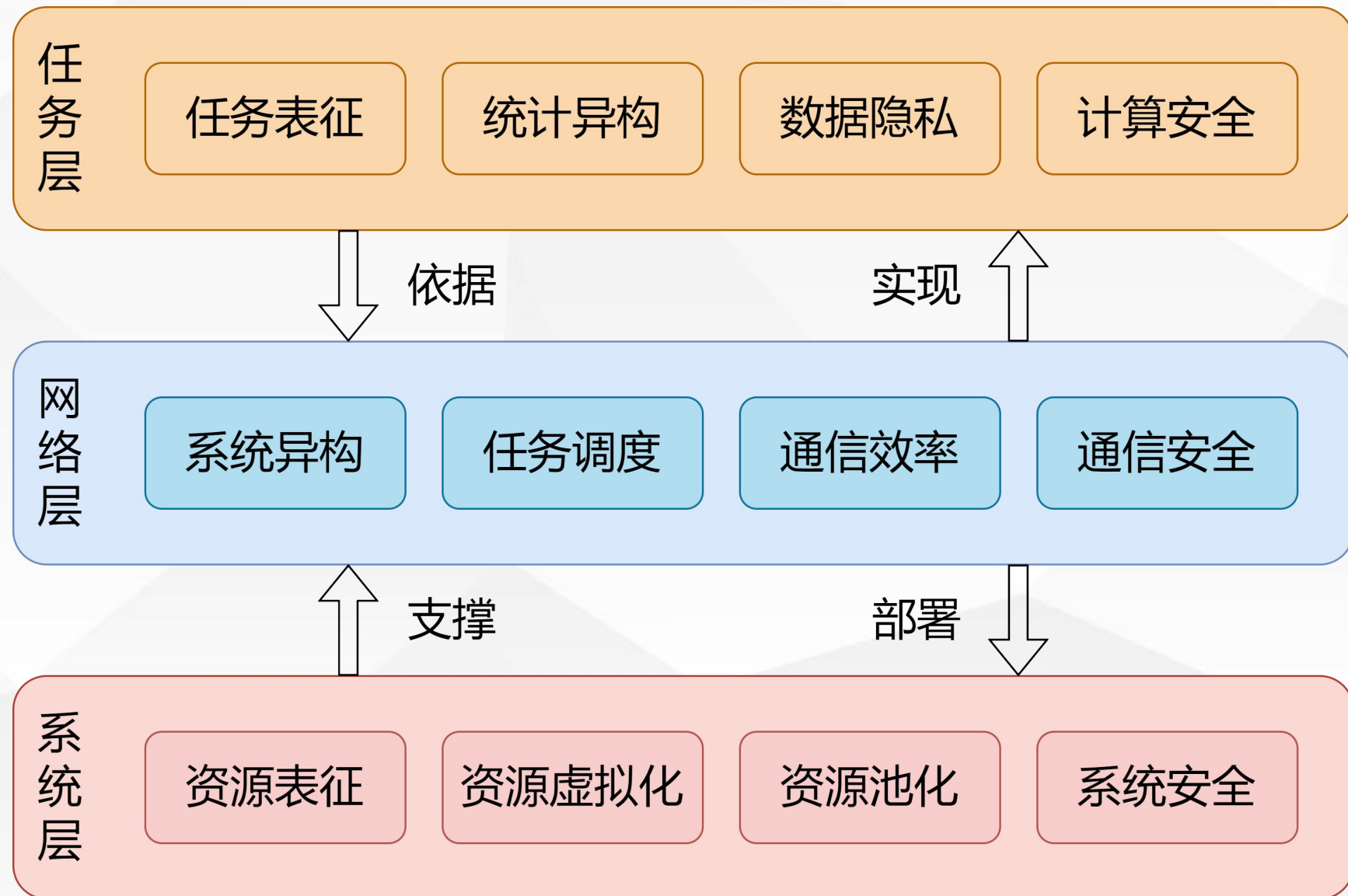
- 明确用户需求
- 应用场景建模

网络层：中层传输角度

- 明确环境限制
- 网络状态建模

系统层：下层资源角度

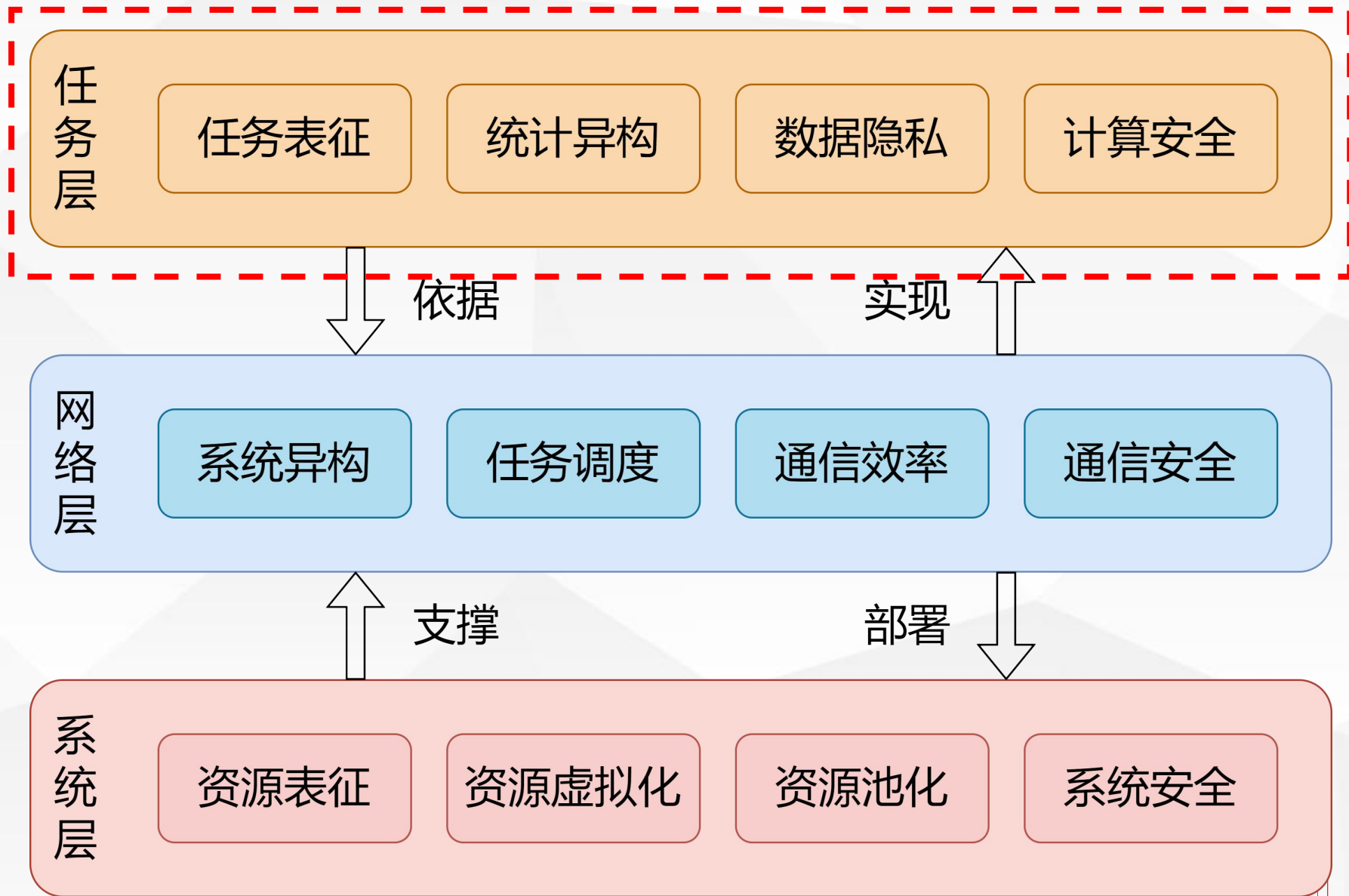
- 明确硬件条件
- 硬件单元建模





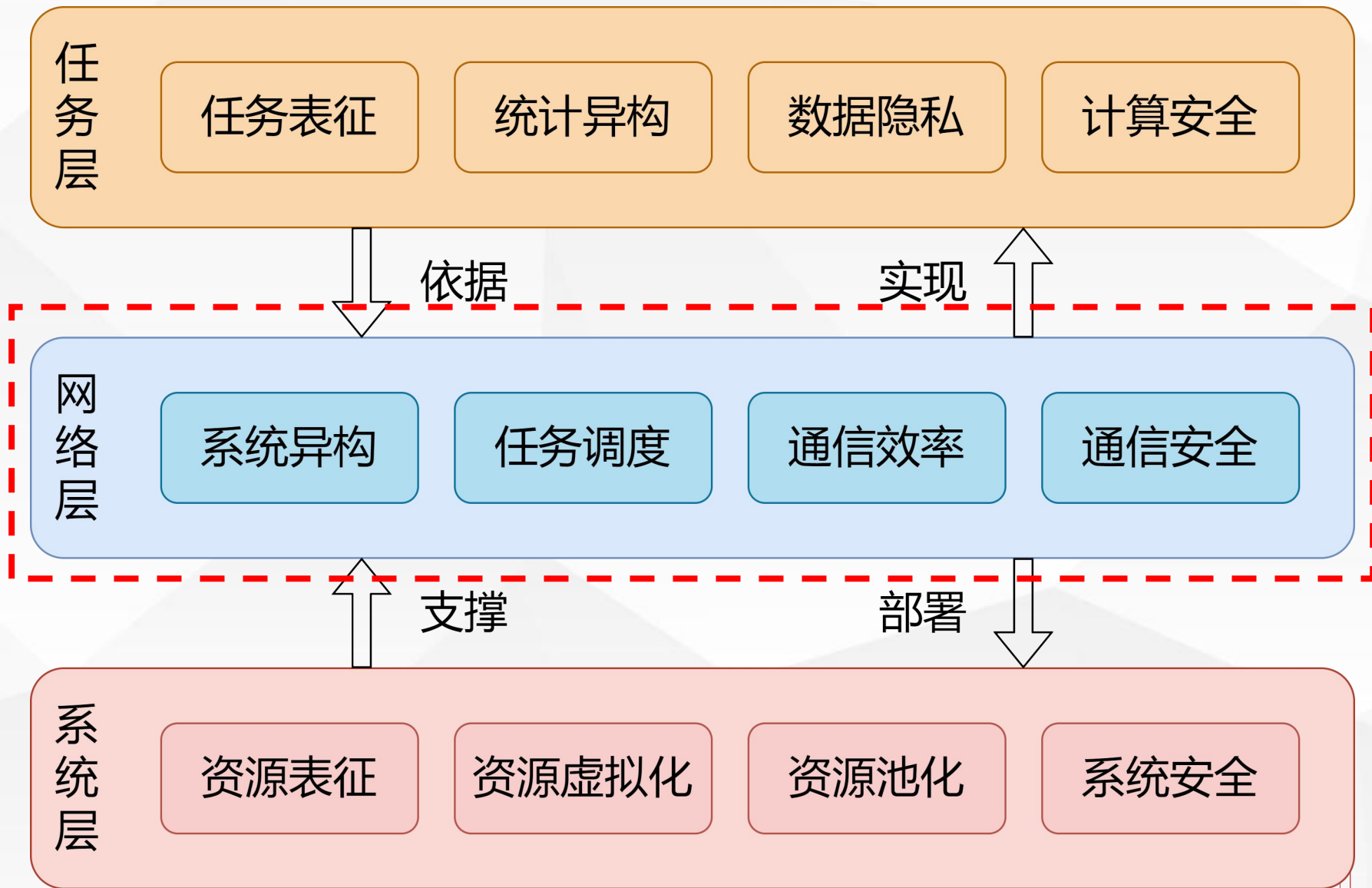
联邦学习挑战-任务层

- ① 上层应用带来的挑战
- ② **任务表征**：如何让系统理解任务需求
- ③ **统计异构**：如何满足不同任务的需求
- ④ **数据隐私**：如何保障任务数据不泄露
- ⑤ **计算安全**：如何保障计算过程正确性





- ❶ 网络传输带来的挑战
- ❷ **系统异构**：如何合理控制不同的设备
- ❸ **任务调度**：如何高效匹配任务和设备
- ❹ **通信效率**：如何减少通信带来的延迟
- ❺ **通信安全**：如何防止通信带来的错误





联邦学习挑战-系统层

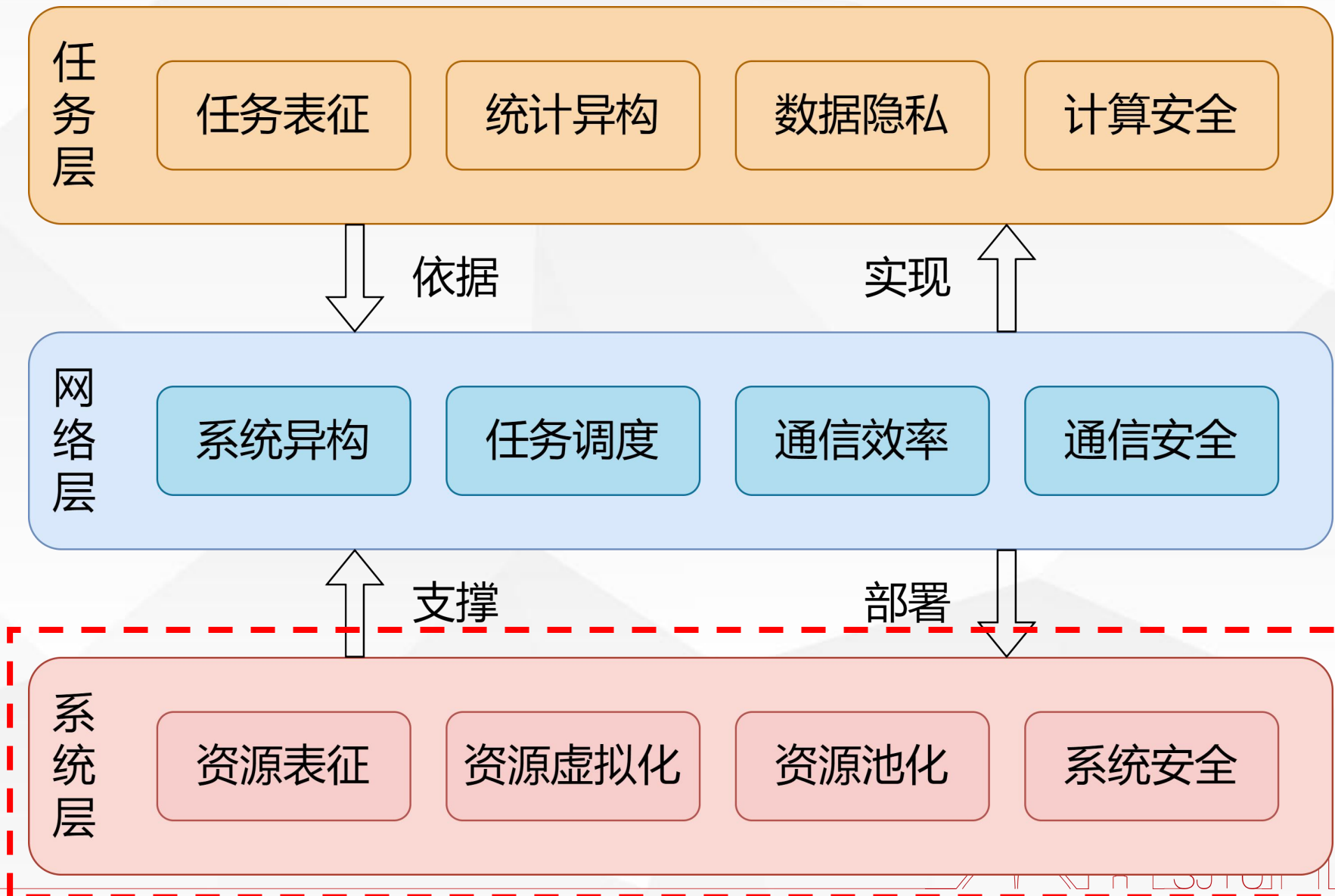
资源角度带来的挑战

资源表征：如何让系统理解资源能力

资源虚拟化：如何让系统调用资源

资源池化：如何让系统统一管理资源

系统安全：如何防止系统受入侵攻击





Relaxing the Core FL Assumptions

完全去中心化/点对点分布式学习

跨数据孤岛的联邦学习

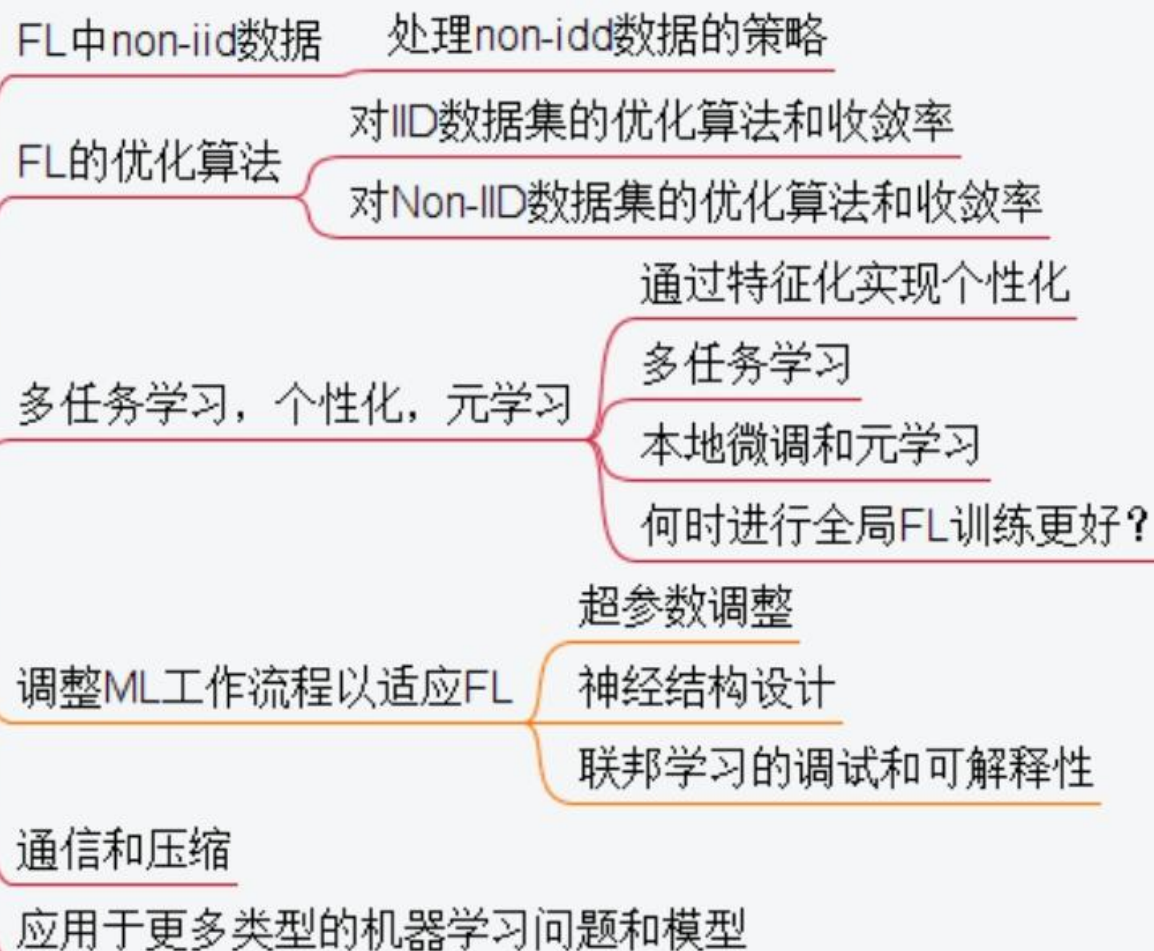
拆分学习(Split Learning)

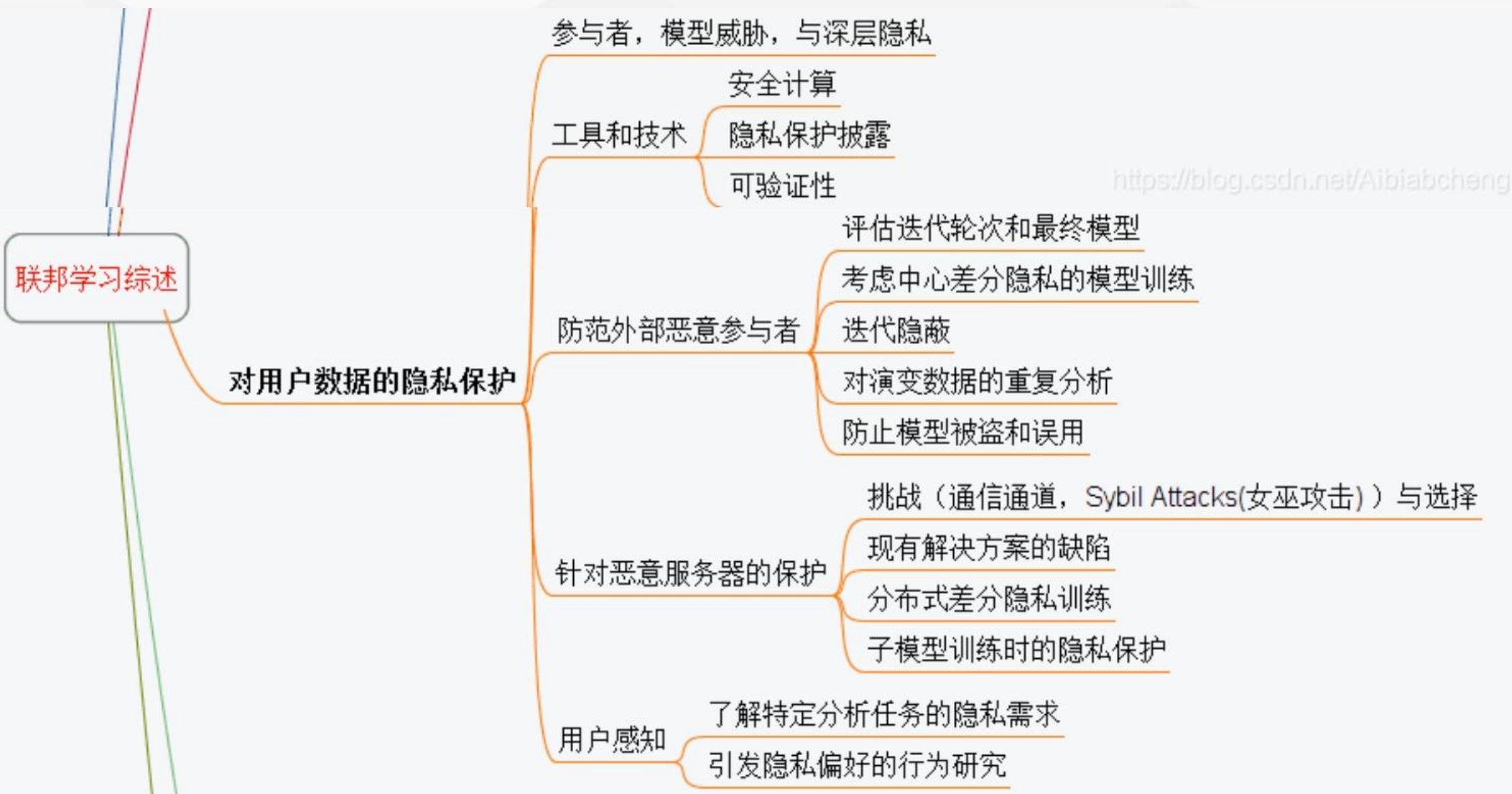
算法挑战

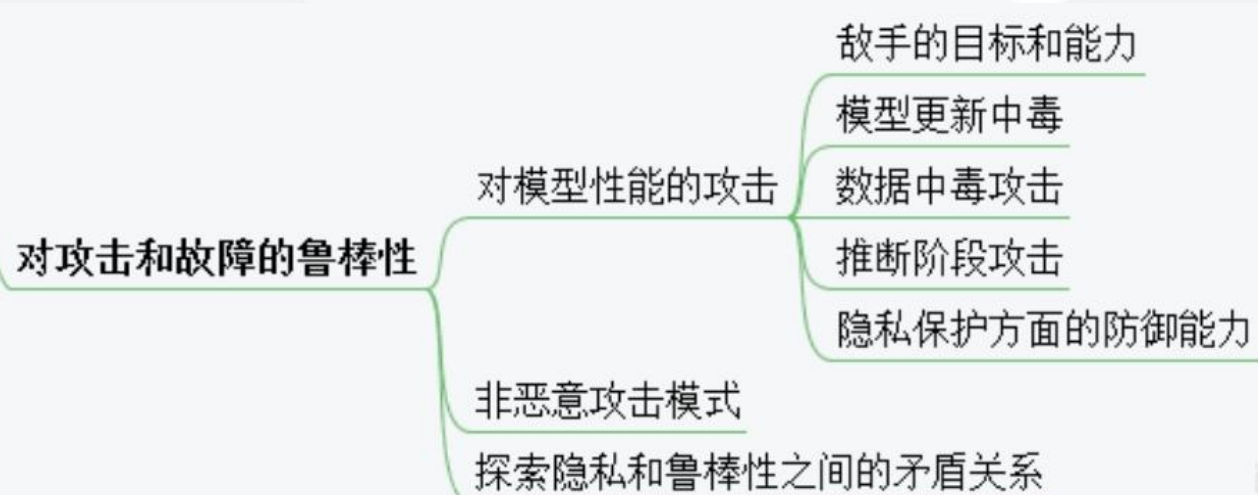
实践挑战



提高效能和效益







<https://blog.csdn.net/Aibiabcheng>



确保公平与解决偏差的来源

训练数据的偏差

不获取敏感属性下的公平性

公平性，隐私性，鲁棒性

通过联邦来提高模型多样性

联邦公平：新机遇和挑战

<https://blog.csdn.net/Aibiabcheng>

01

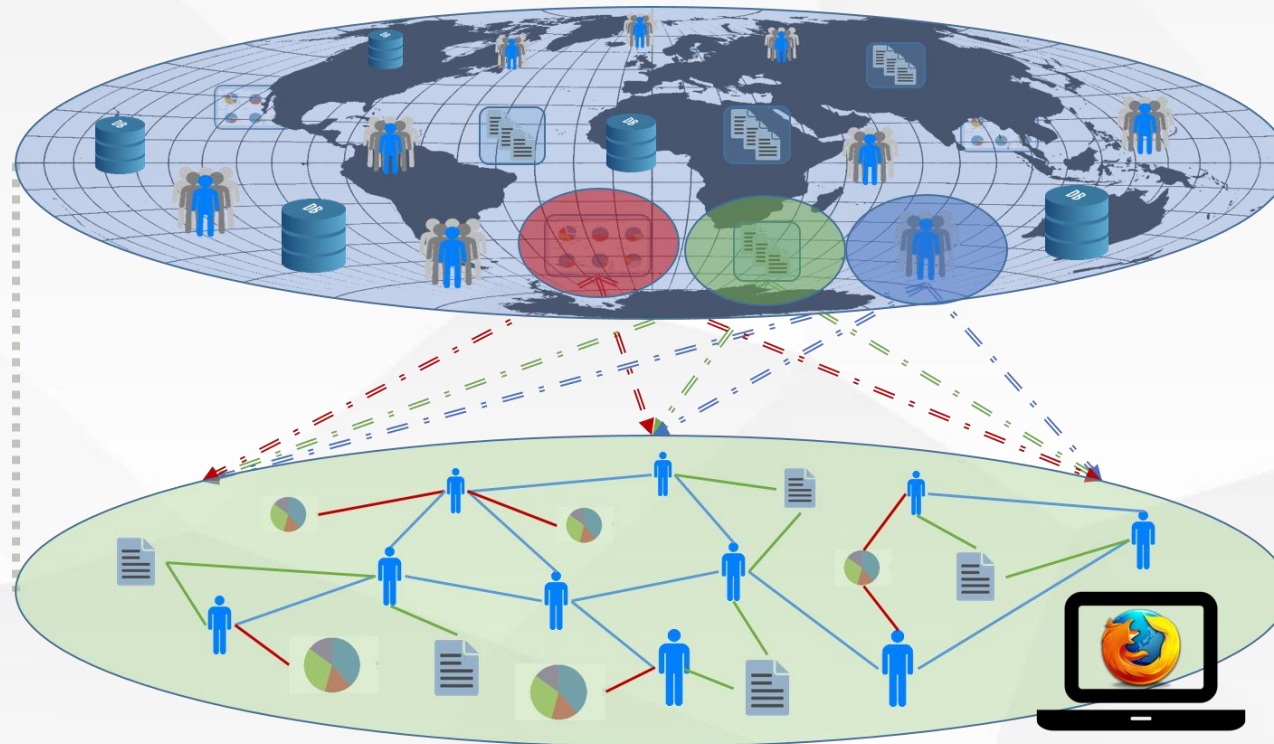
Challenges in FL

- Expensive communication
- Systems heterogeneity
- Statistical heterogeneity
- Privacy concerns

Challenges in FL

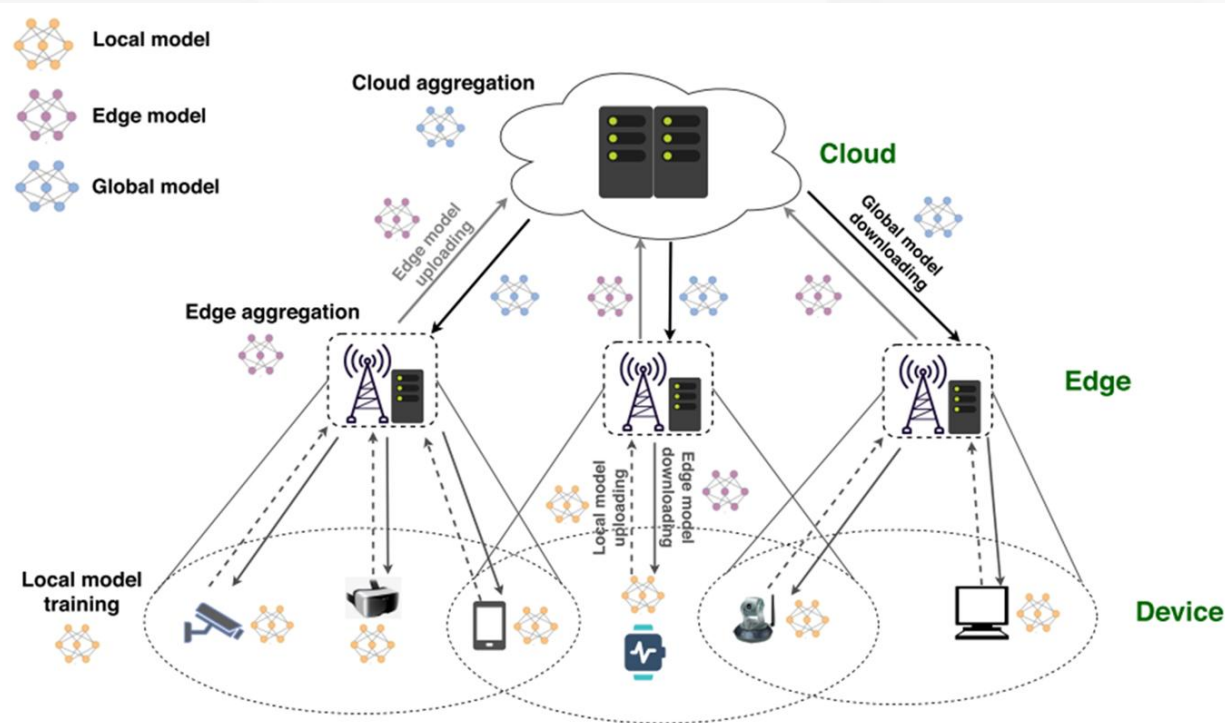
⦿ Expensive Communication.

- federated networks are potentially comprised of a massive number of devices, e.g., millions of smart phones, and communication in the network can be slower than local computation by many orders of magnitude.



Systems Heterogeneity.

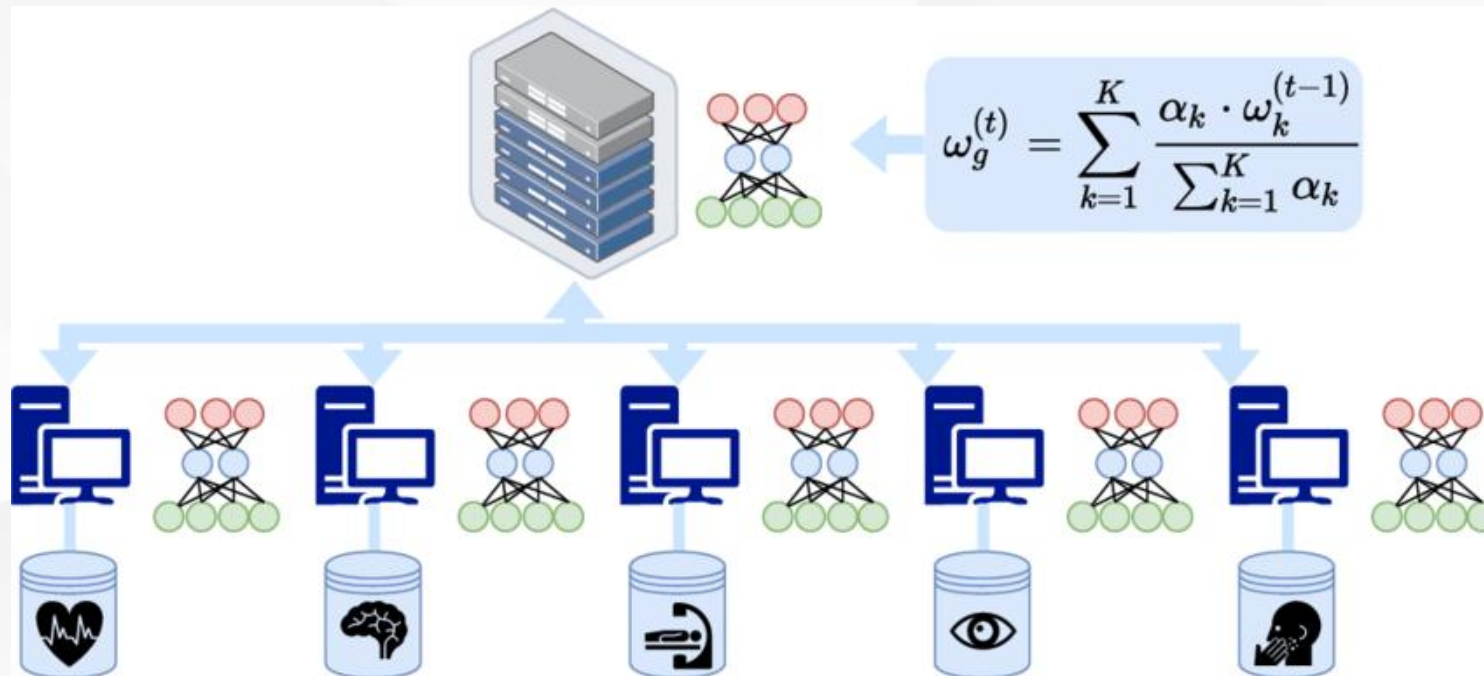
- The storage, computational, and communication capabilities of each device in federated networks may differ due to variability in hardware (CPU, memory), network connectivity (3G, 4G, 5G, wifi), and power (battery level).
- Each device may also be unreliable.



Challenges in FL

Statistical Heterogeneity.

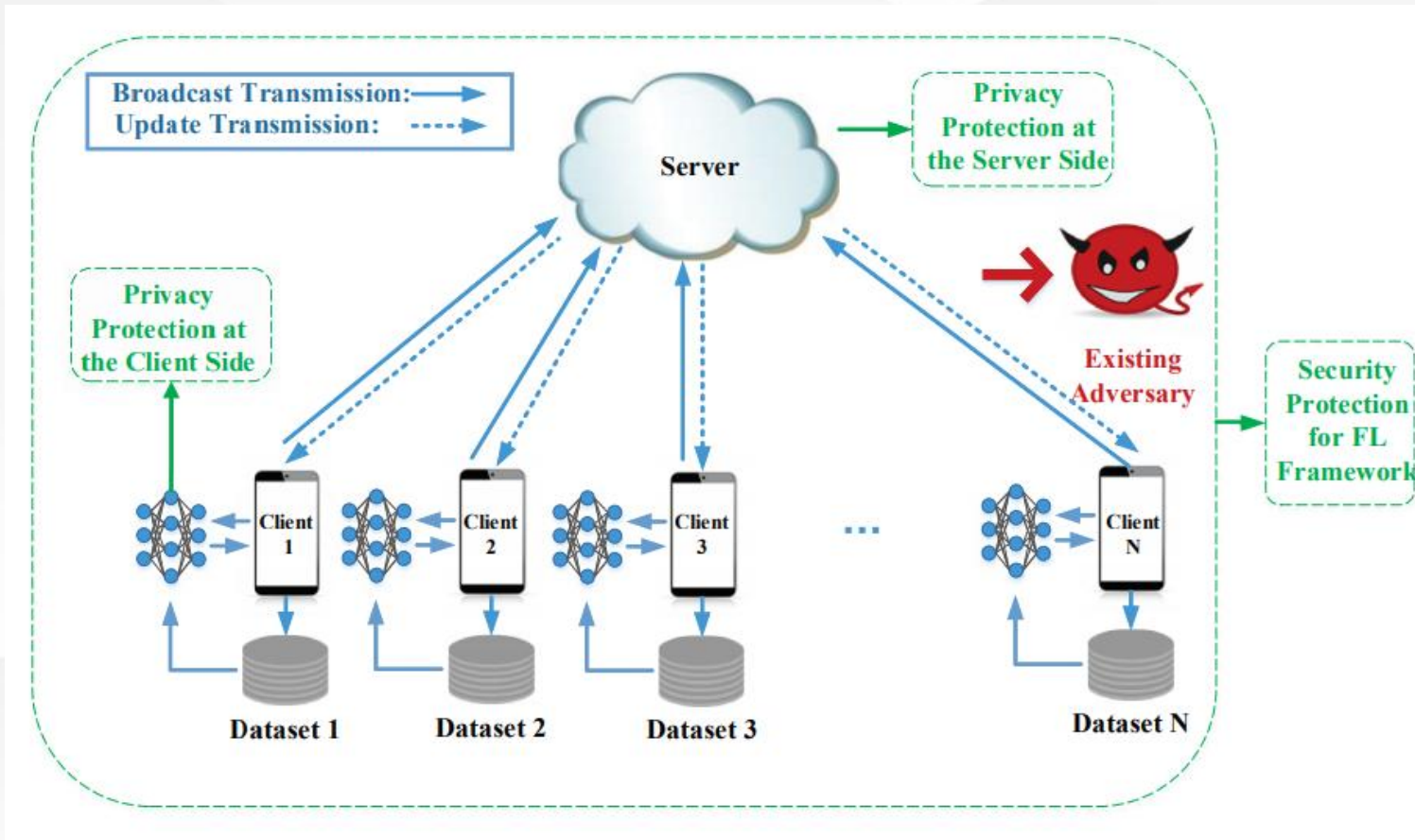
- Devices frequently generate and collect data in a non-identically distributed manner across the network, e.g., mobile phone users have varied use of language in the context of a next word prediction task.
- Increases the likelihood of stragglers.



Challenges in FL

Privacy Concerns.

- communicating model updates throughout the training process can nonetheless reveal sensitive information





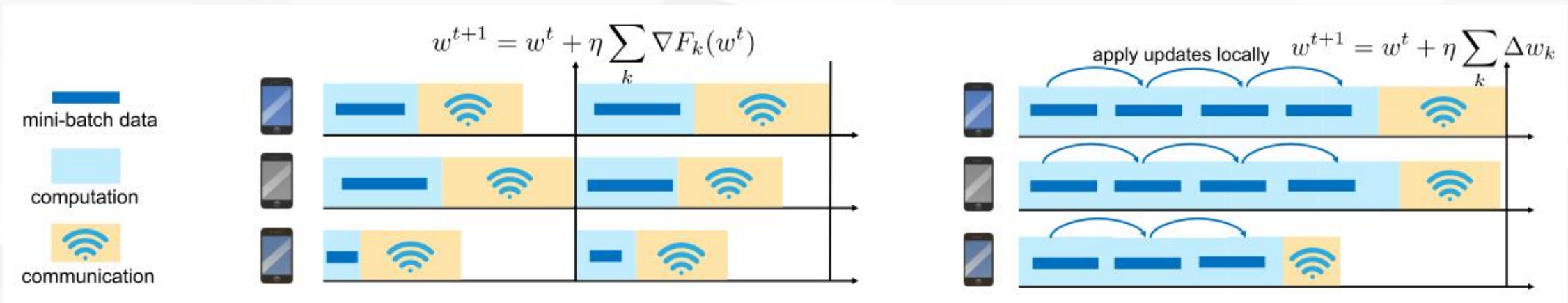
02

Communication-efficiency

- Local updating
- Compression schemes
- Decentralized training

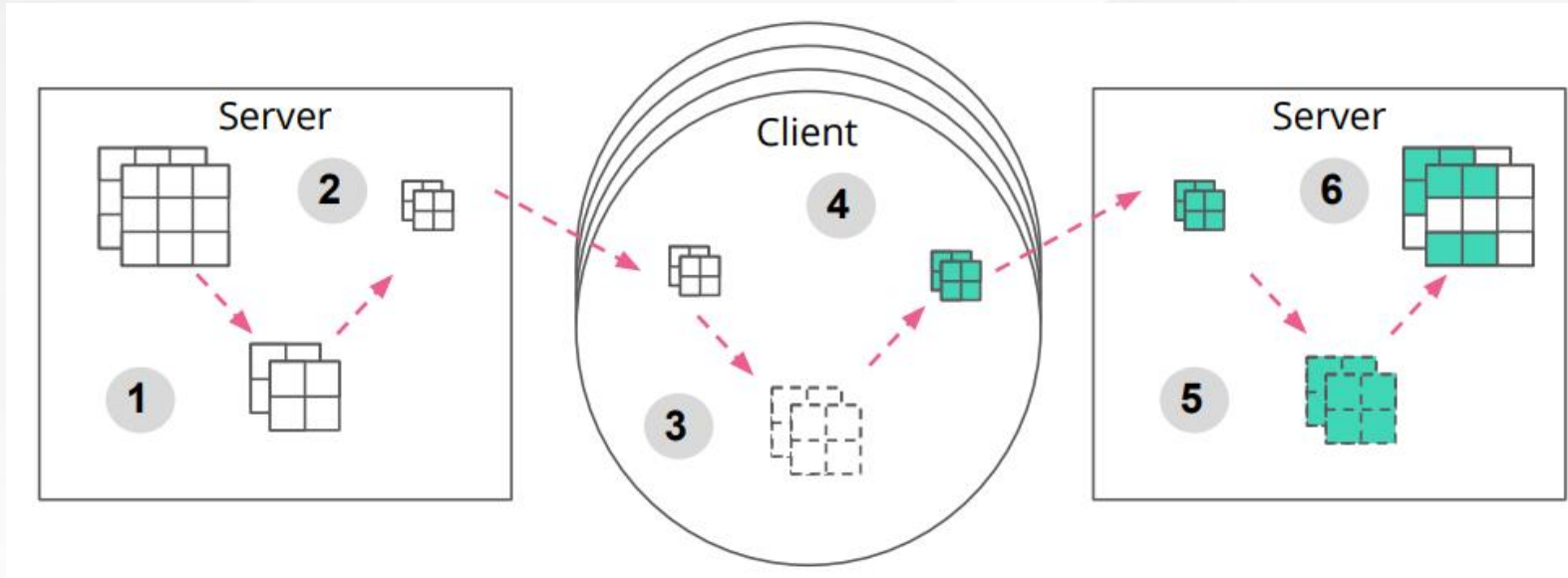
Local updating

- Mini-batch optimization methods have been shown to have limited flexibility to adapt to communication-computation trade-offs that would maximally leverage distributed data processing.
- Allow for a variable number of local updates to be applied on each machine in parallel at each communication round
- For convex objectives, distributed local-updating primal-dual methods have emerged as a popular way to tackle such a problem.



Compression schemes

- Use lossy compression and dropout to reduce server-to-device communication.



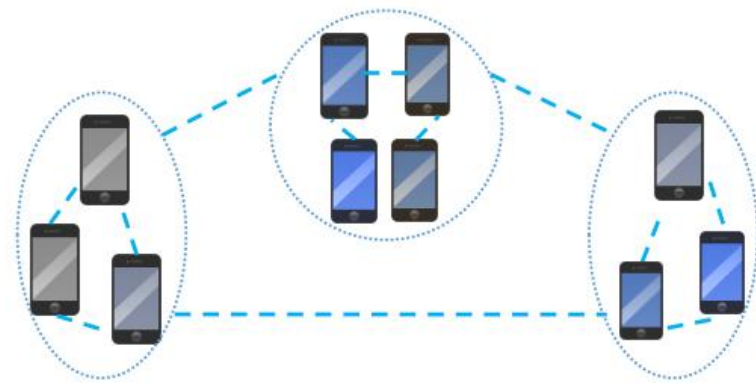
(1) constructing a sub-model via Federated Dropout, and by (2) lossily compressing the resulting object. This compressed model is then sent to the client, who (3) decompresses and trains it using local data, and (4) compresses the final update. This update is sent back to the server, where it is (5) decompressed and finally, (6) aggregated into the global model

Decentralized Training

- In federated learning, a star network (where a central server is connected to a network of devices) is the predominant communication topology.
- Decentralized algorithms can in theory reduce the high communication cost on the central server.



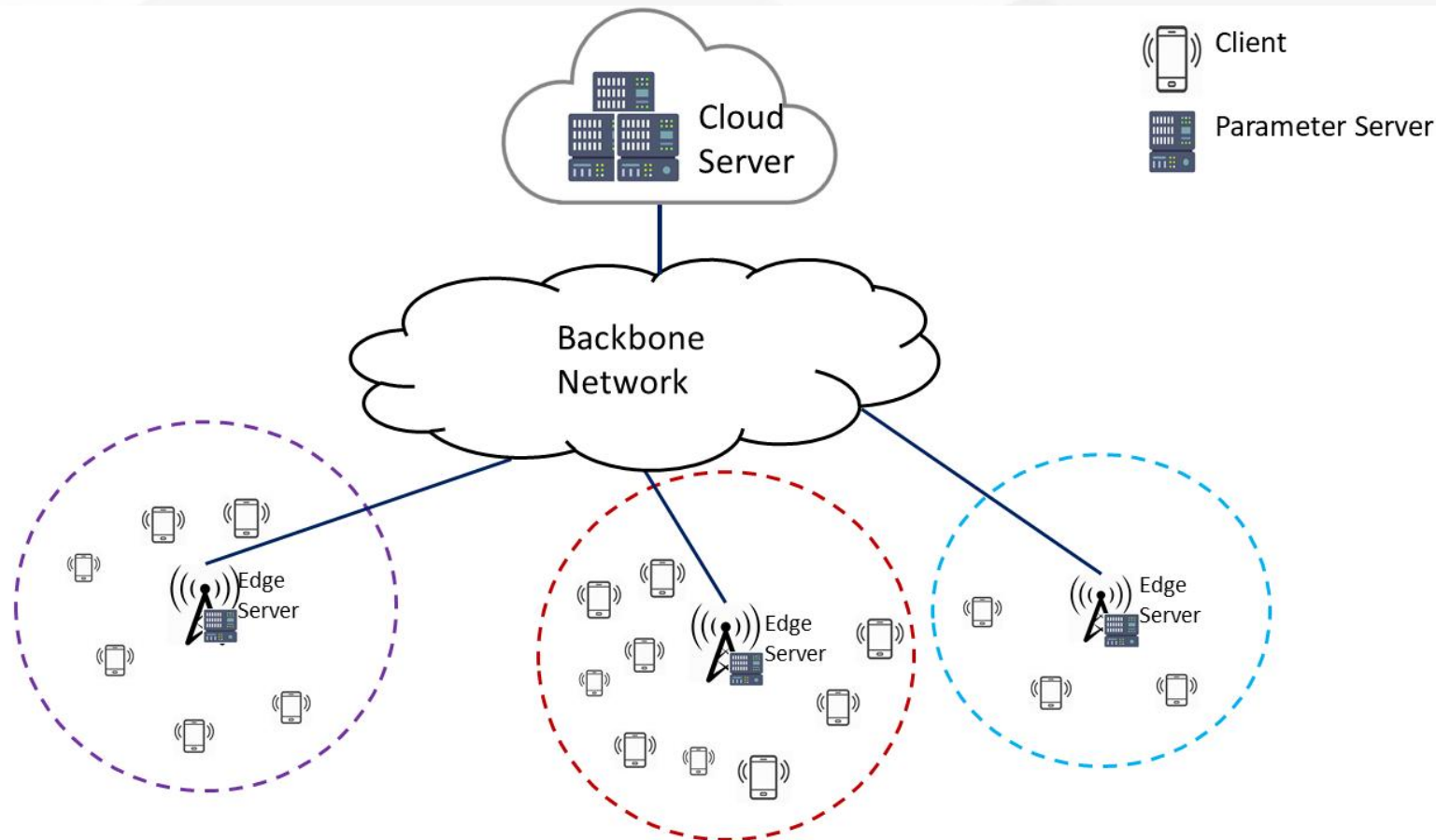
Centralized topology



Decentralized topology

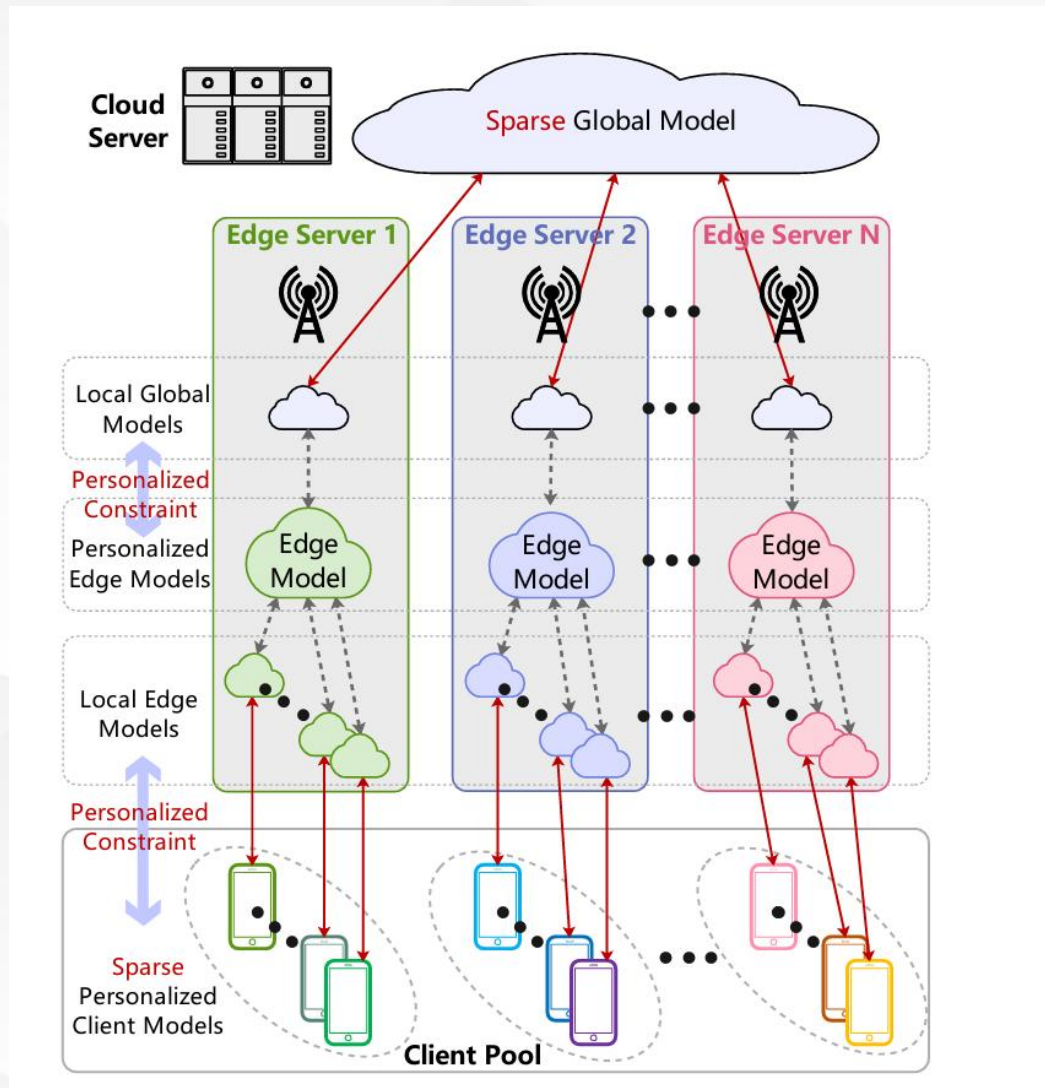
Decentralized Training

- Hierarchical communication patterns.



④ Sparse Gradient Compression

- Top-k gradient selection with error feedback, reducing communication by 90% while maintaining accuracy.
- Example: SparseFL algorithm achieves 1.2x faster convergence than FedAvg.



03

Privacy protection

- Privacy threats/attacks in federated learning (FL)
- Enhance the general privacy-preserving feature of FL
- Associated cost of the privacy-preserving techniques

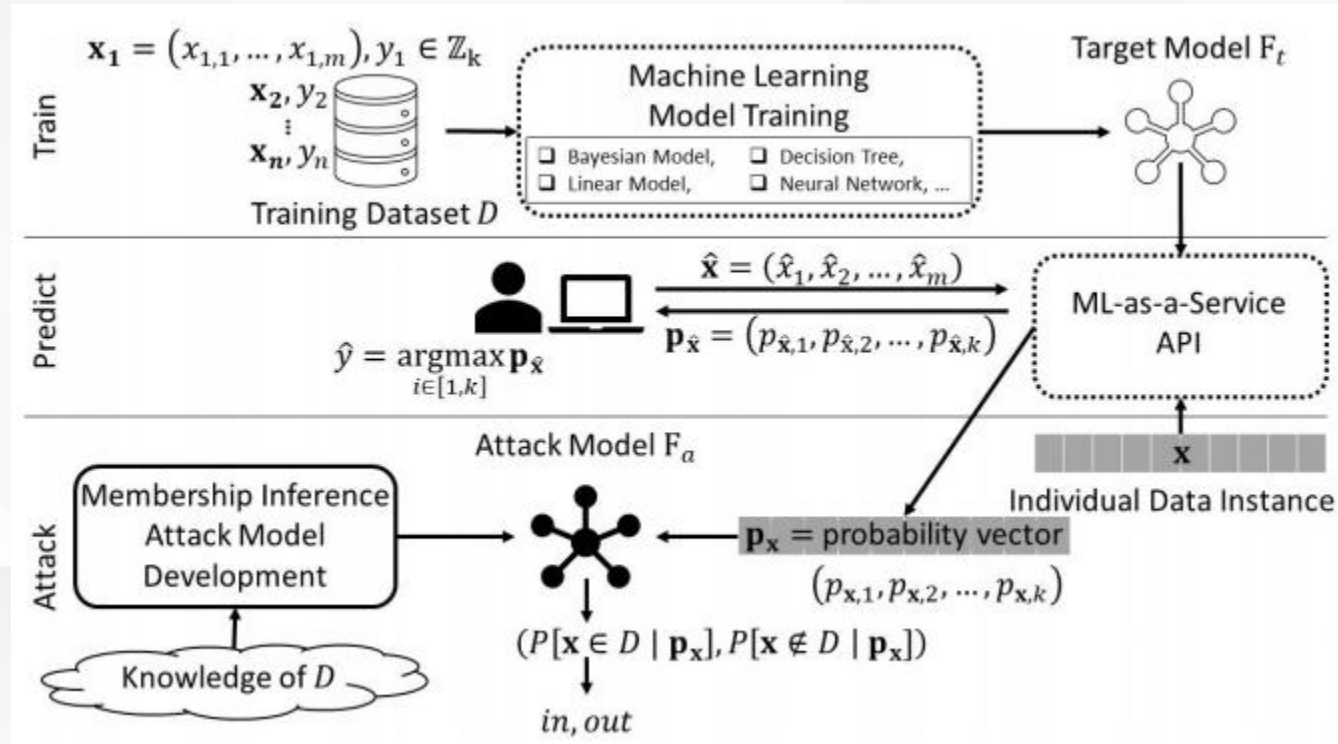


Privacy threats/attacks in FL

- Membership inference attacks
- Unintentional data leakage and reconstruction through inference

Membership inference attacks

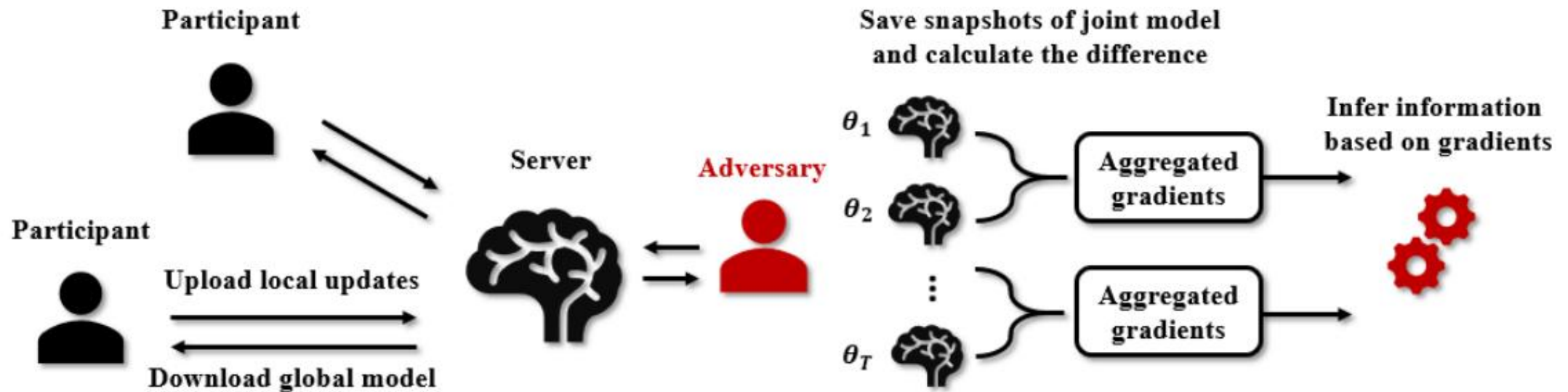
- The neural network is vulnerable to memorize their training data which is prone to passive and active inference attacks.
- The attacker misuses the global model to get information on the training data of the other users.



Privacy protection

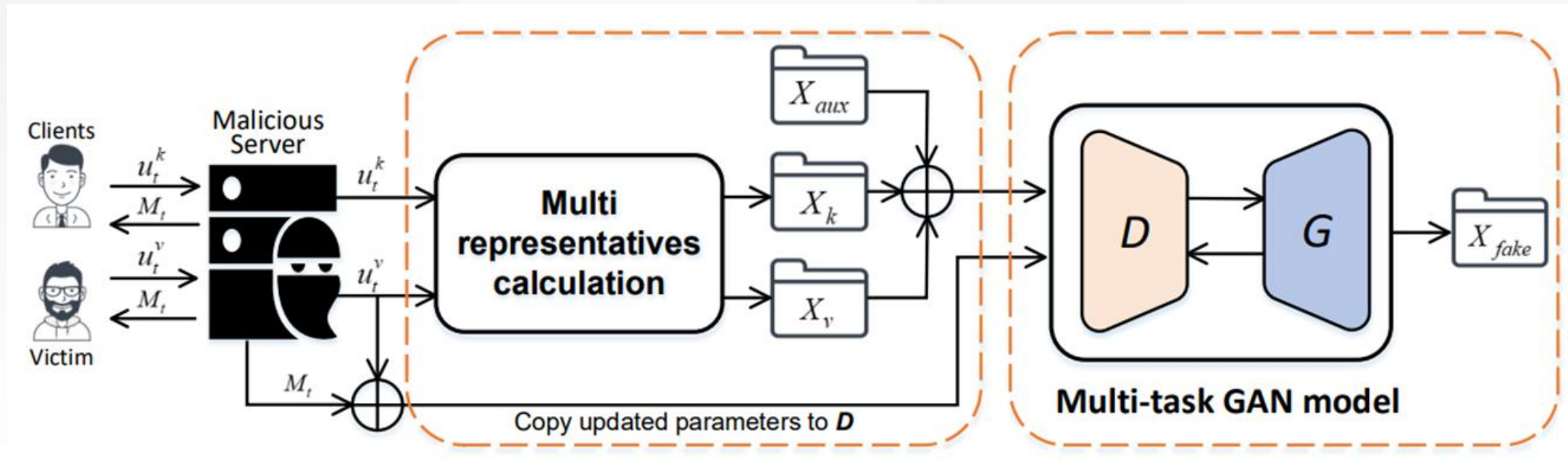
Unintentional data leakage and reconstruction

- Is a scenario where updates or gradients from clients leak unintended information at the central server.



GANs-based inference attacks

- GANs are generative adversarial networks that have gained much popularity in big data domains.
- It is possible to have potential adversaries among FL clients.





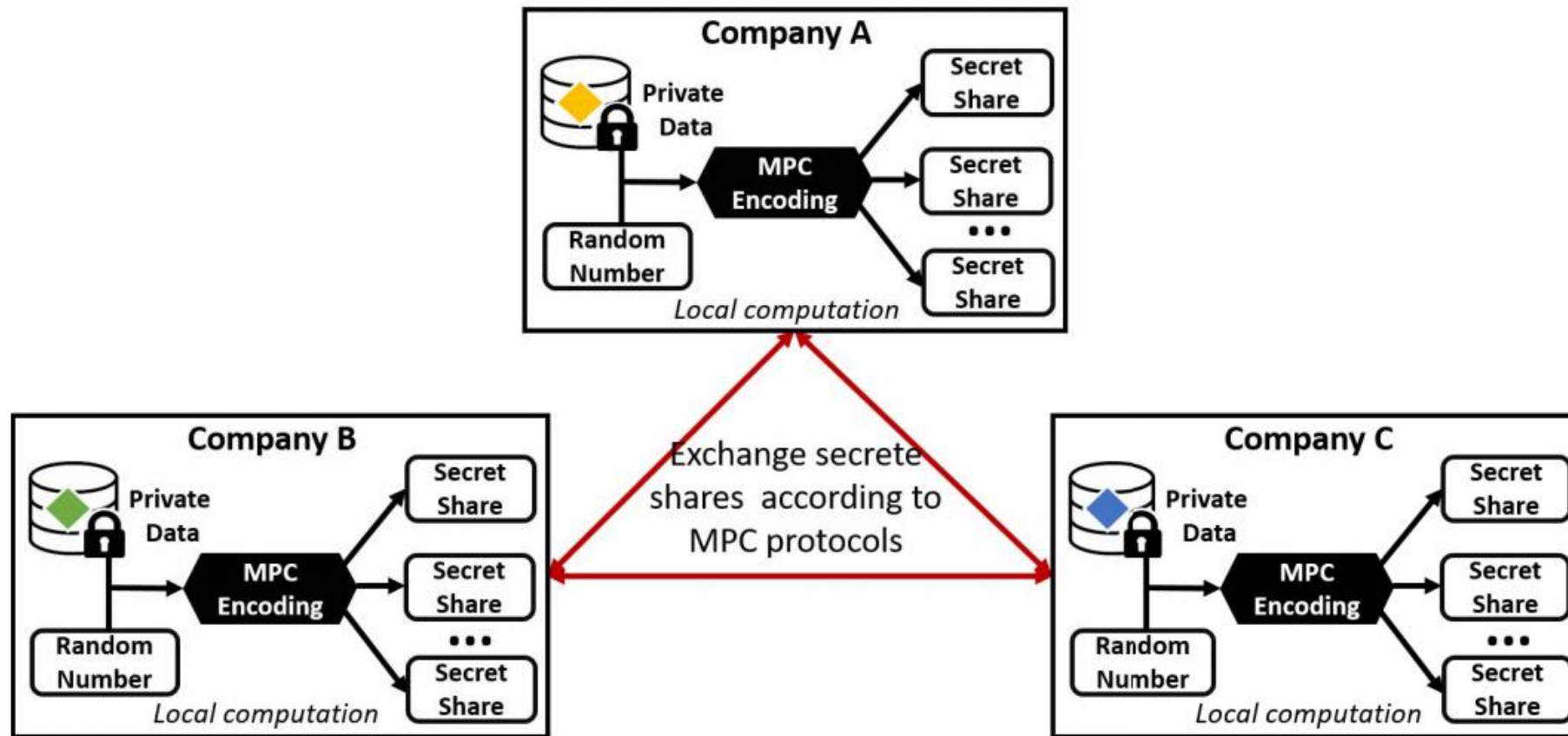
Enhance privacy-preserving in FL

- Secure multi-party computation
- Differential privacy
- VerifyNet
- Adversarial training

Privacy protection

Secure multi-party computation

- Secure the inputs of multi-participant while they jointly compute a model or a function.

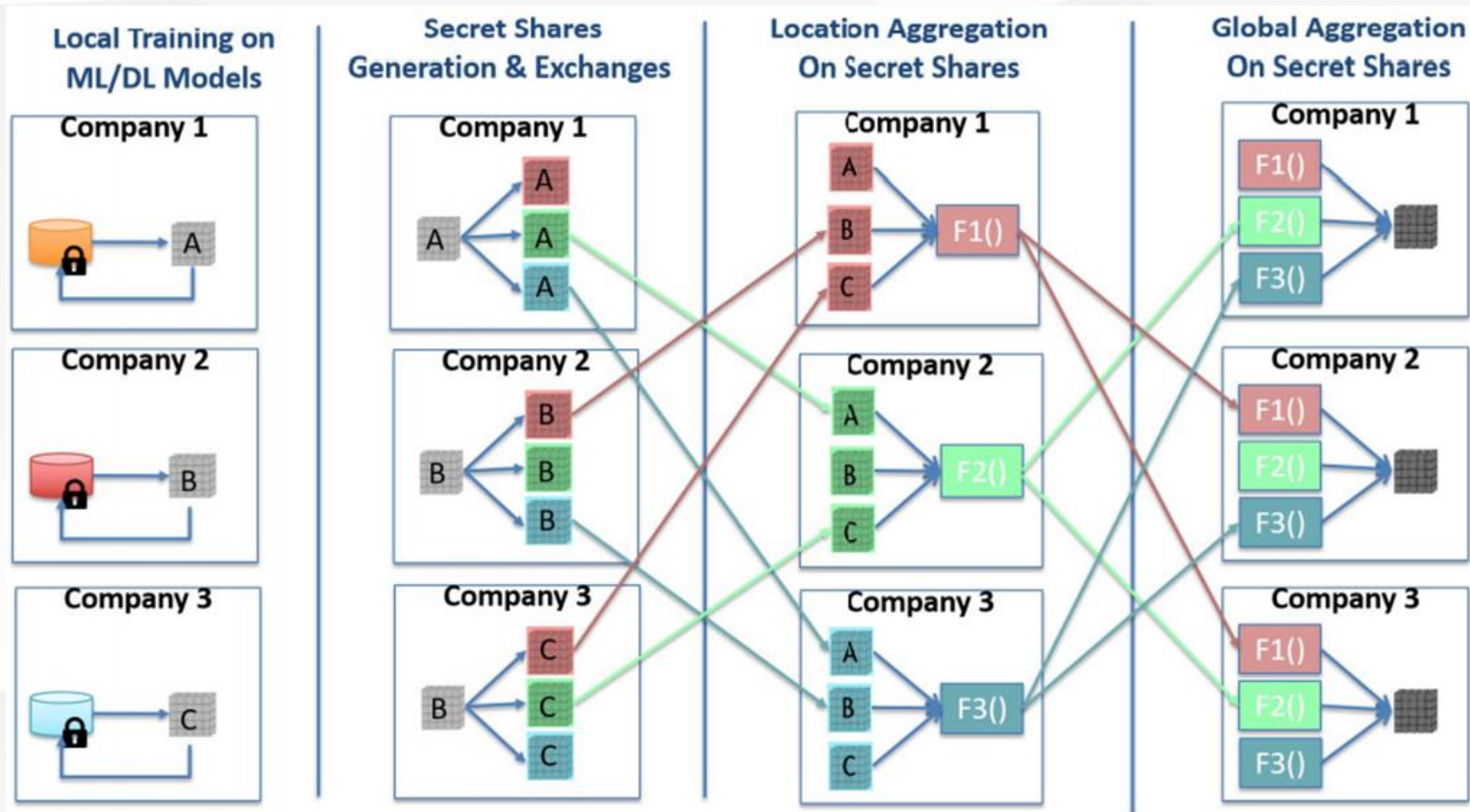


Privacy protection



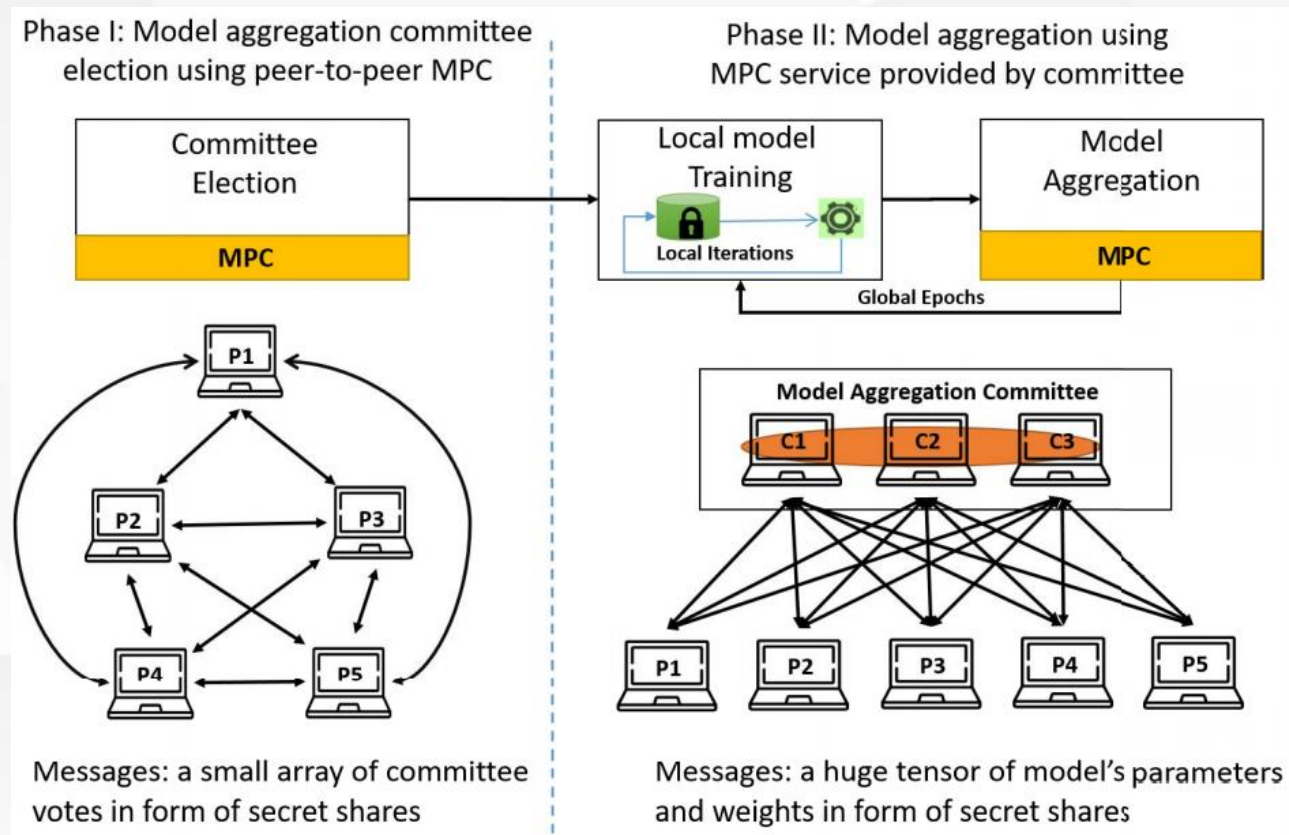
Secure multi-party computation

- Secure the inputs of multi-participant while they jointly compute a model or a function.



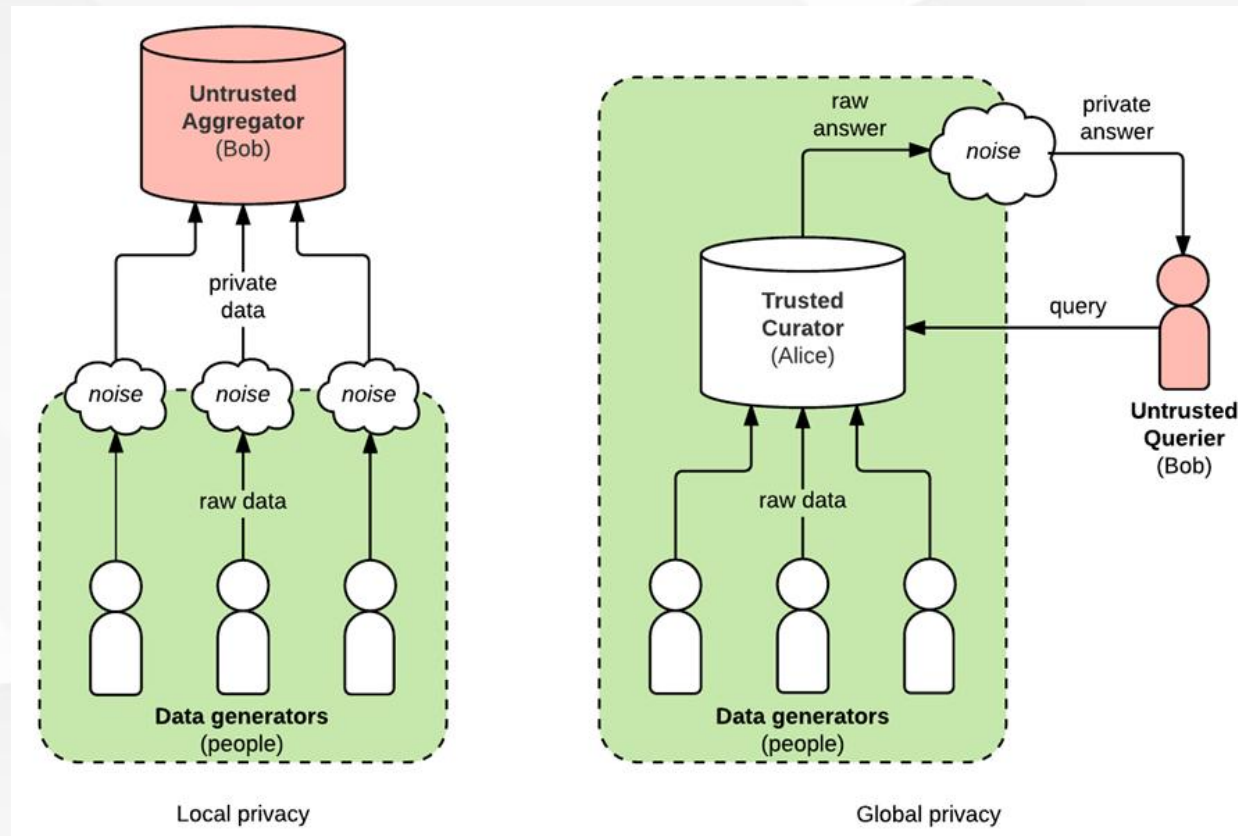
Secure multi-party computation

- In FL, the computing efficiency is increased immensely since it only needs to encrypt the parameters instead of the large volume of data inputs.



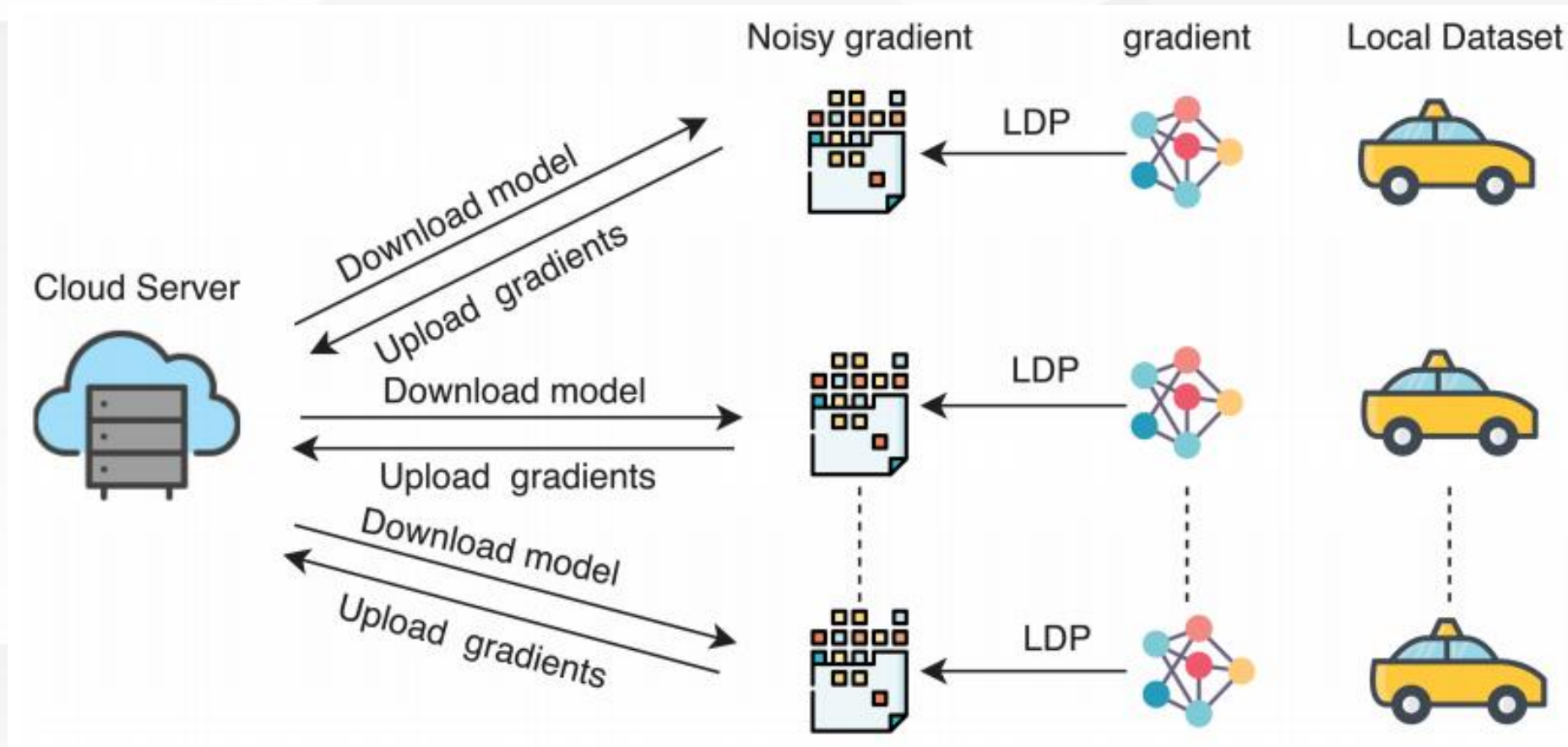
Differential Privacy

- Add noise to personal sensitive attributes



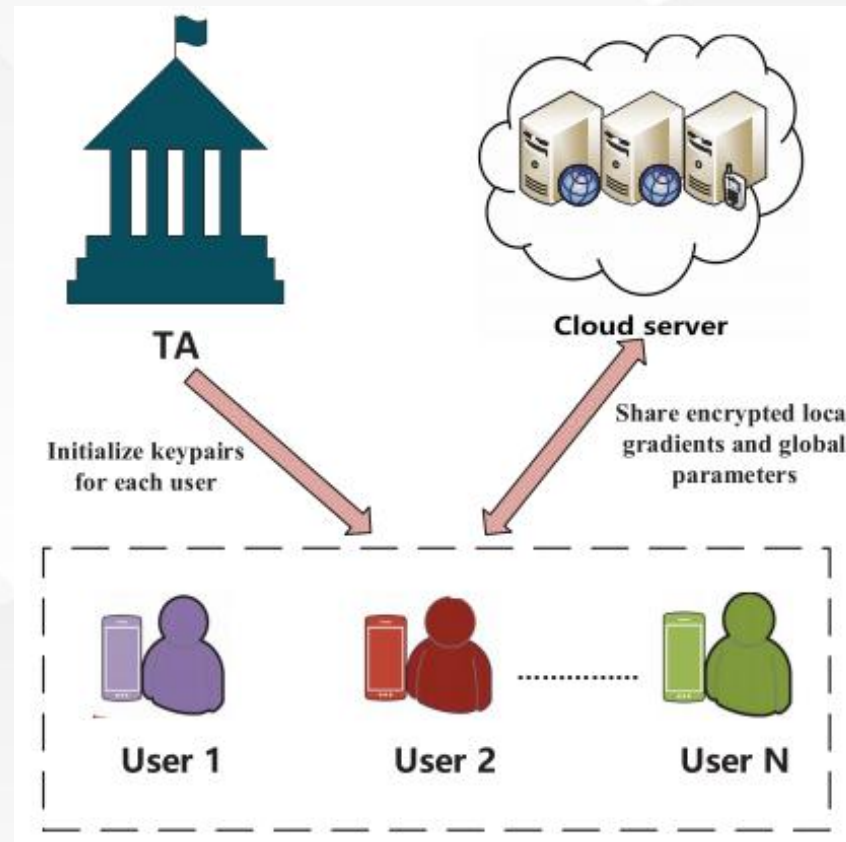
Differential Privacy

- DP is introduced to add noise to participants' uploaded parameters



VerifyNet

- It gets listed as a preferred mitigation strategy to preserve privacy as it provides double-masking protocol which makes it difficult for attackers to infer training data.





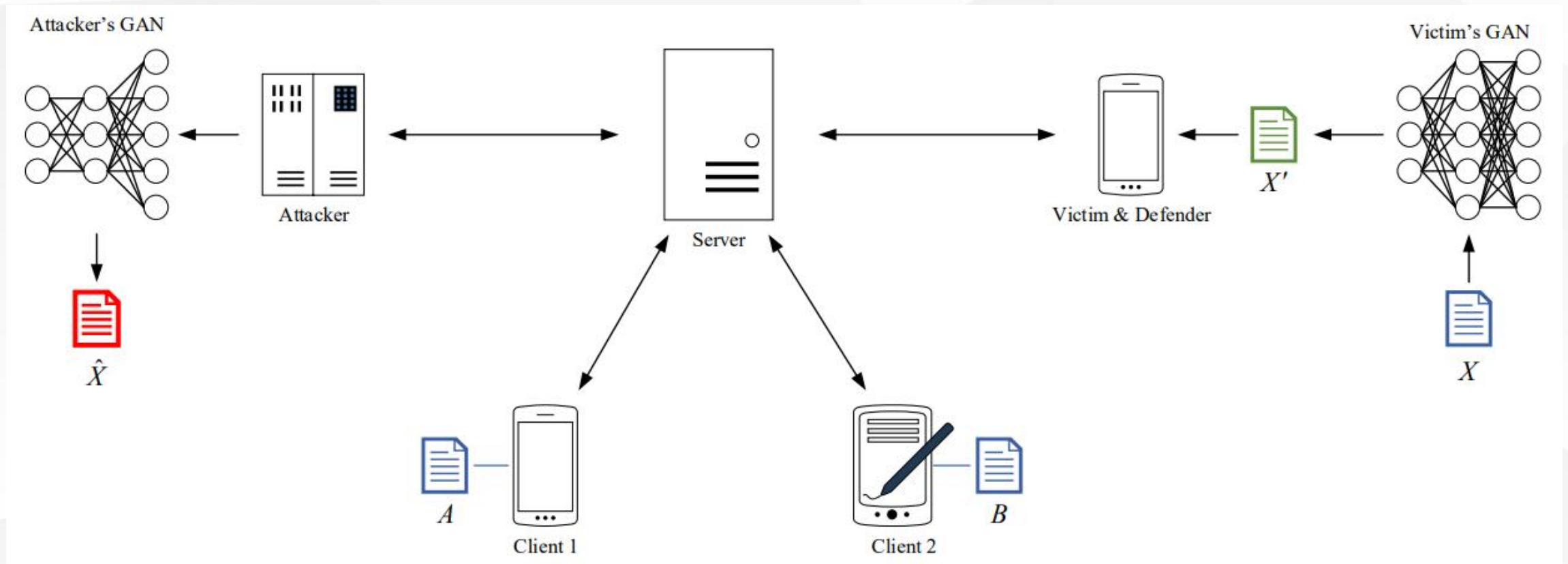
Adversarial training

- Evasion attacks from an adversarial user aims to fool ML models by injecting adversarial samples into the machine learning models.
- The attacker tries to impact the robustness of the FL model with perturbed data.
- Adversarial training, which is a proactive defense technique, tries all permutations of an attack from the beginning of the training phase to make the FL global model robust to known adversarial attacks.

Privacy protection

Adversarial training

- Use GAN to generate fake training data.





Privacy protection

Associated cost

Approach	Cost	Methodology
Secure Multi-party Computation	Efficiency loss due to encryption	Encrypt uploaded parameters
Differential Privacy	Accuracy loss due to added noise in client's model	Add random noise to uploaded parameters
Hybrid	Subdued cost on both efficiency and accuracy	Encrypt the manipulated parameter
VerifyNet	Communication overhead	Double-masking protocol Verifiable aggregation results
Adversarial Training	Computation power, training time for adversarial samples	Include adversarial samples in training data

04

Heterogeneity

- Addressing systems heterogeneity
- Addressing statistical heterogeneity



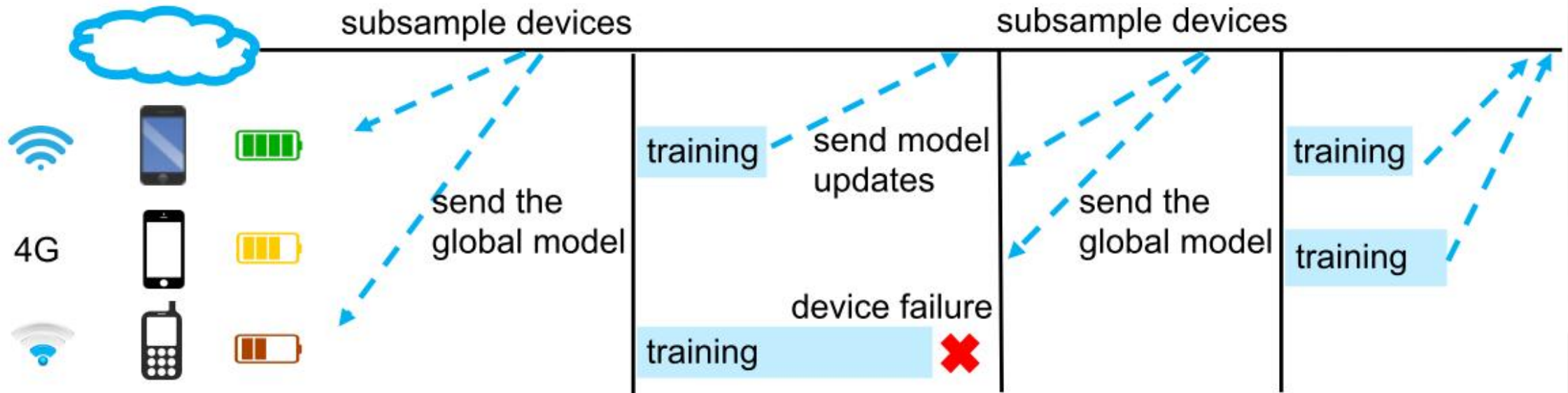
Addressing systems heterogeneity

- Asynchronous communication
- Active sampling
- Fault tolerance
- Using client-specific model

Heterogeneity

Asynchronous communication

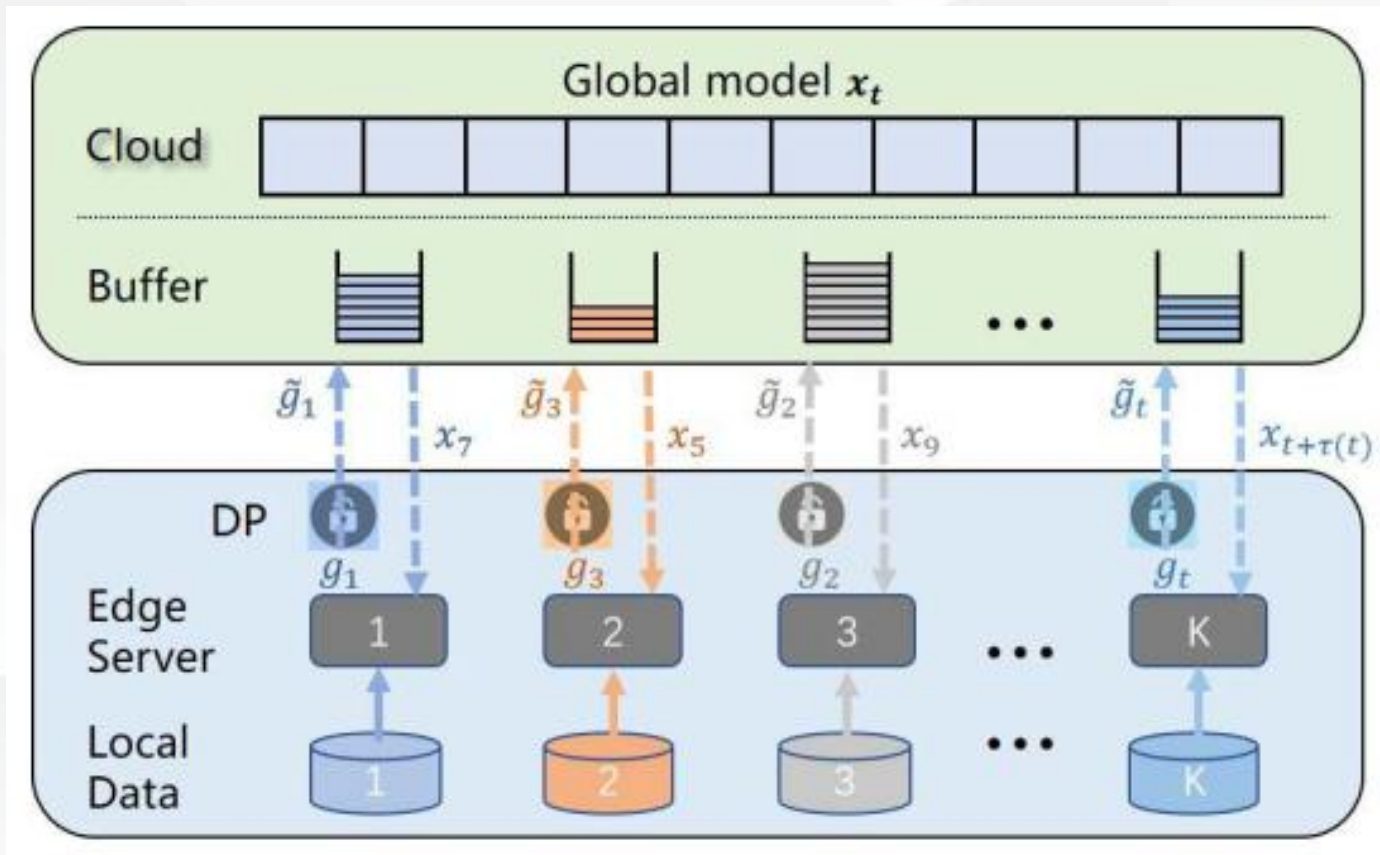
- Synchronous schemes are simple and guarantee a serial-equivalent computational model, but they are also more susceptible to stragglers in the face of device variability.



Heterogeneity

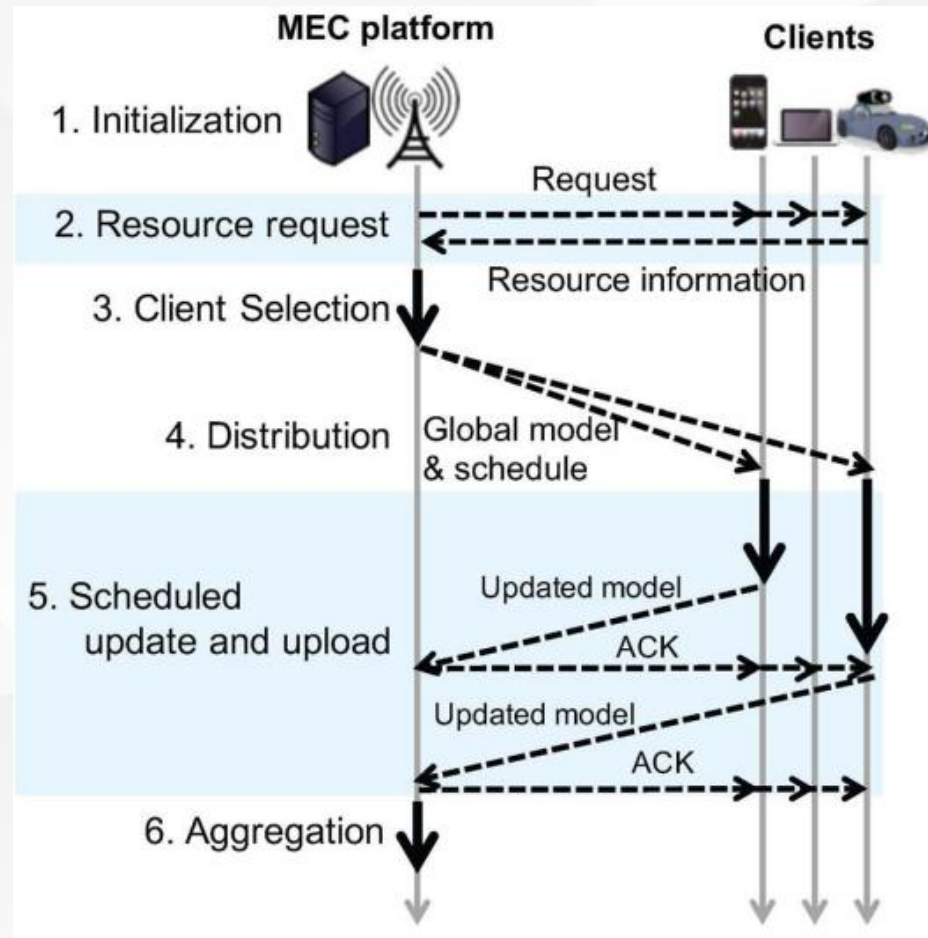
Asynchronous communication

- Asynchronous schemes are an attractive approach to mitigate stragglers in heterogeneous environments.



Active sampling

- Actively selecting participating devices at each round.



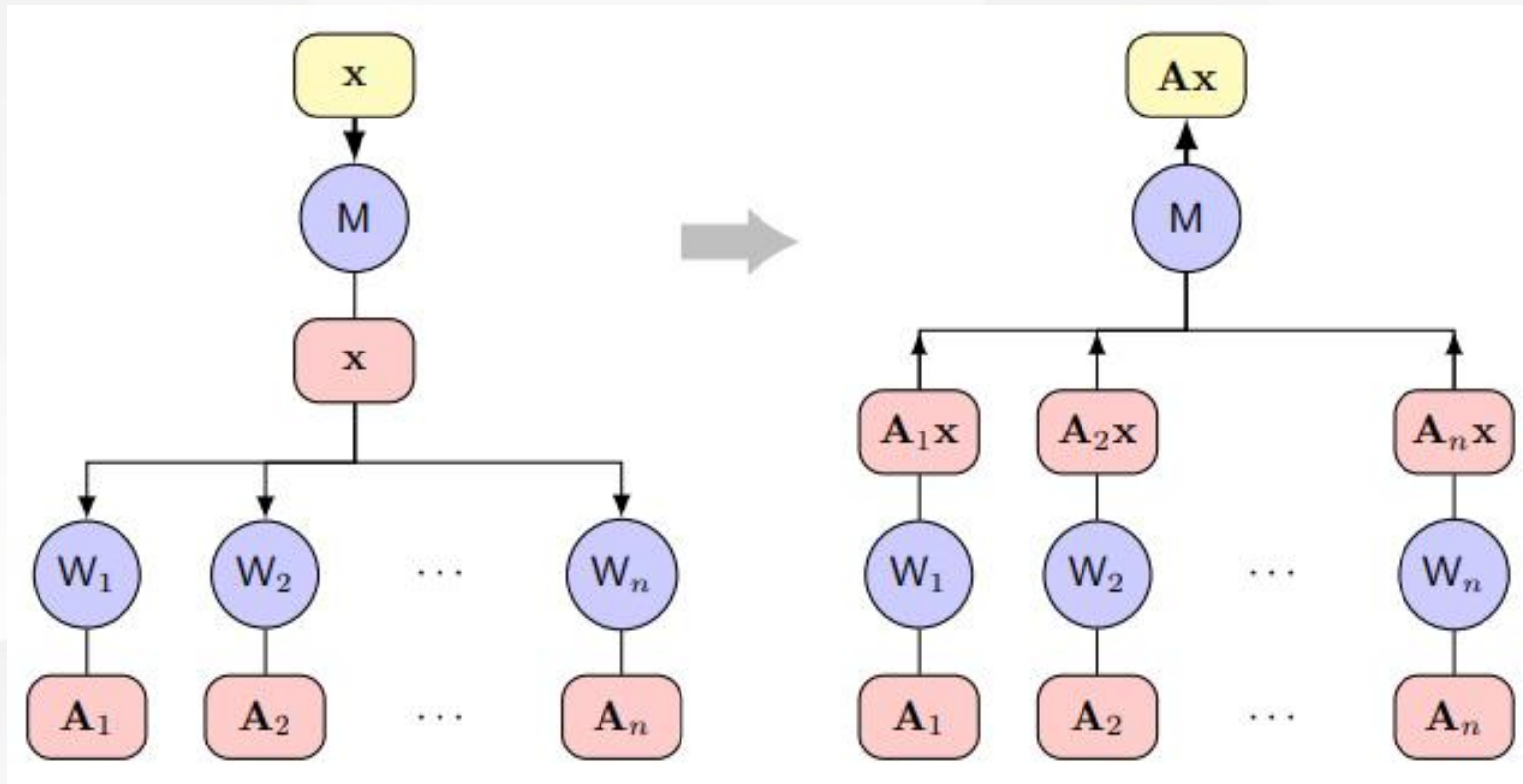


Fault tolerance

- Fault tolerance has been extensively studied in the systems community and is a fundamental consideration of classical distributed systems.
- When learning over remote devices, however, fault tolerance becomes more critical.
- One practical strategy is to simply ignore such device failure, which may introduce bias into the device sampling scheme if the failed devices have specific data characteristics.

Fault tolerance

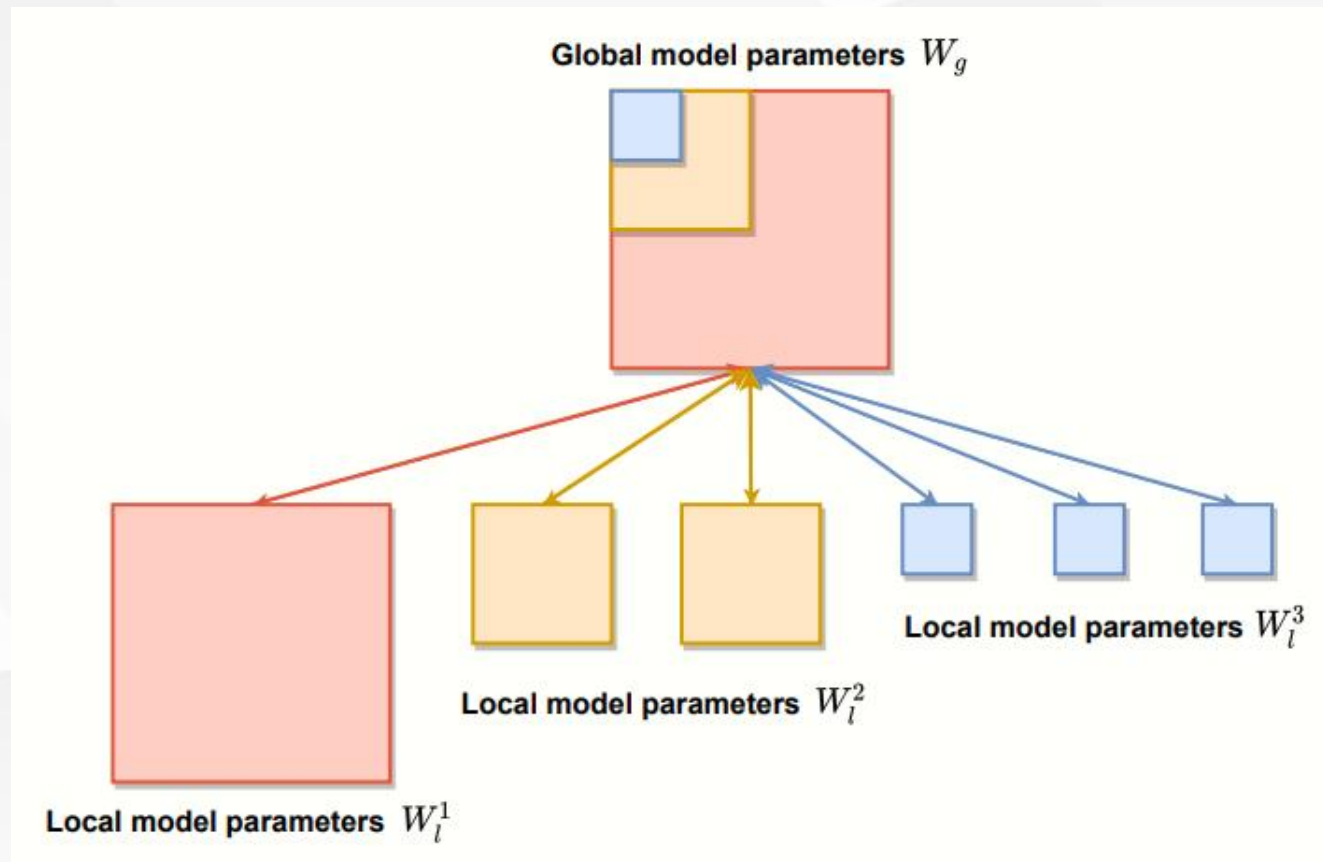
- Coded computation is another option to tolerate device failures by introducing algorithmic redundancy.



Heterogeneity

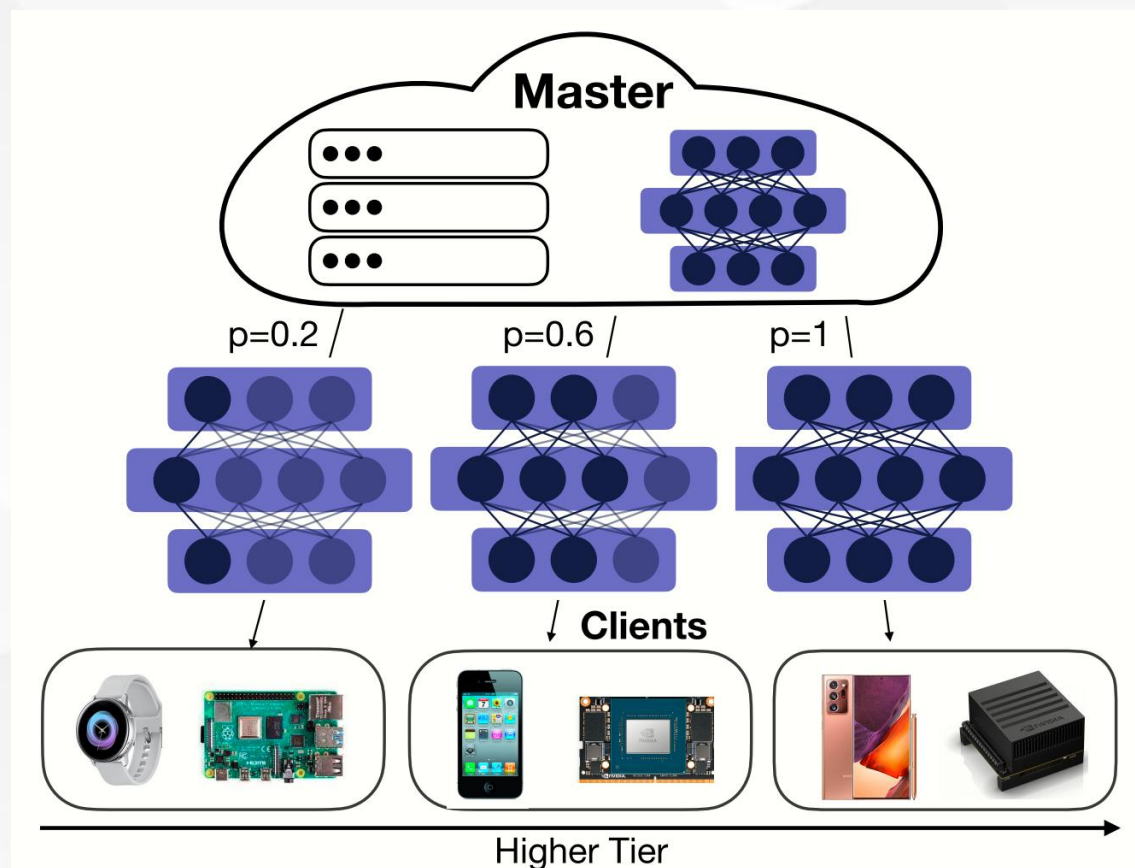
Using client-specific model

- HeteroFL trains heterogeneous local models and aggregate them stably and effectively into a single global inference model.



Using client-specific model

- FjORD employs Ordered Dropout (OD) to tailor the amount of computation to the capabilities of each participating device.

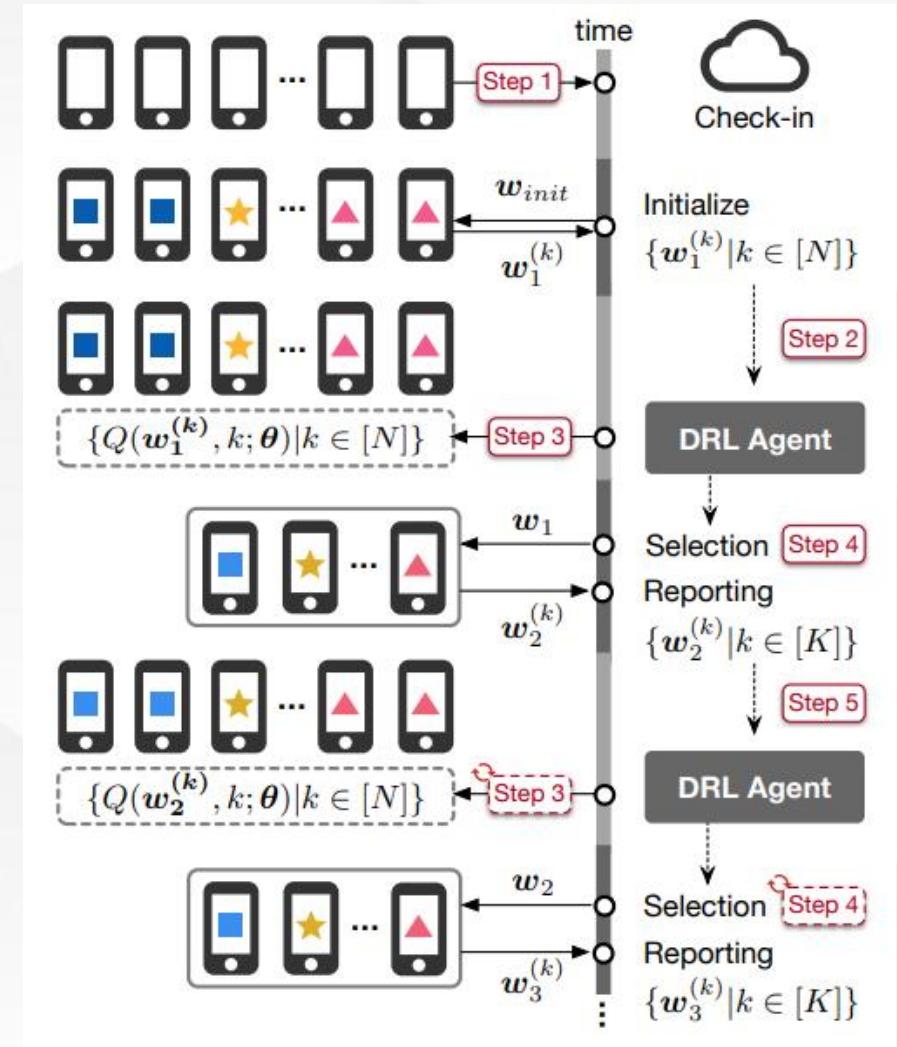




- **Statistical heterogeneity**
 - Overcome the non-IID and unbalanced issue
 - Utilize the non-IID and unbalanced characteristic



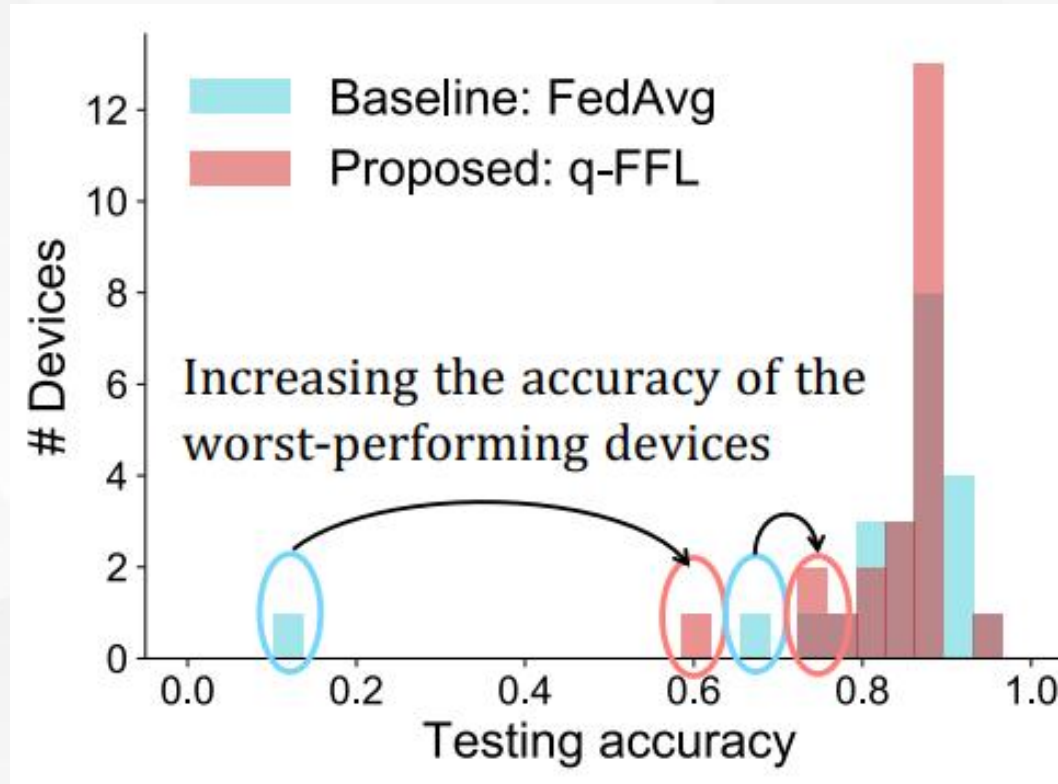
- Overcome the non-IID and unbalanced issue
 - Although the data is not independent and identically distributed among all the clients, we can relieve this issue by client selection.
 - Client selection can be formulated as a deep reinforcement learning problem in federated learning.
 - It solely relies on model weight information to determine which device may improve the global model the most —thus preserving the same level of privacy as the original FL does.





Overcome the non-IID and unbalanced issue

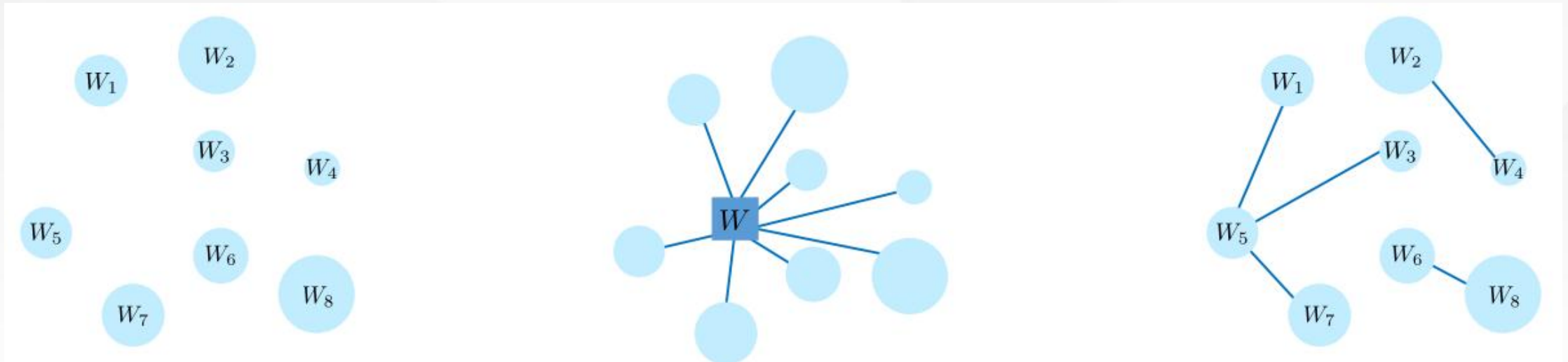
- Devices with higher loss are given higher relative weight to encourage less variance in the final accuracy distribution.



Heterogeneity

Utilize the non-IID and unbalanced characteristic

- Non-IID data is not just an issue for federated learning, but also a natural feature in this setting.
- Personalized federated learning is welcomed.



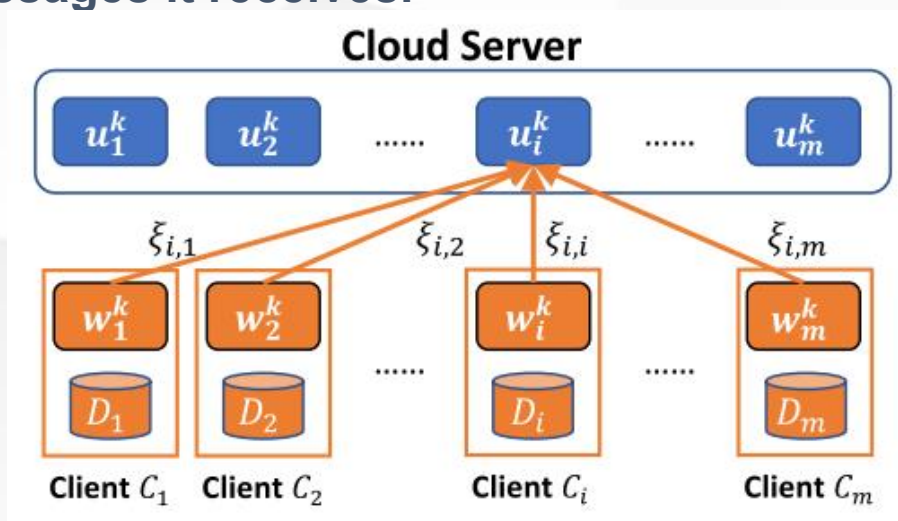
Learn personalized models for each device; do not learn from peers.

Learn a global model; learn from peers.

Learn personalized models for each device; learn from peers.

Personalized federated learning

- FedAMP allows each client to own a local personalized model, it maintains a personalized cloud model on the cloud server for each client.
- FedAMP realizes the attentive message passing mechanism by attentively passing the personalized model of each client as a message to the personalized cloud models with similar model parameters.
- FedAMP updates the personalized cloud model of each client by a weighted convex combination of all the messages it receives.

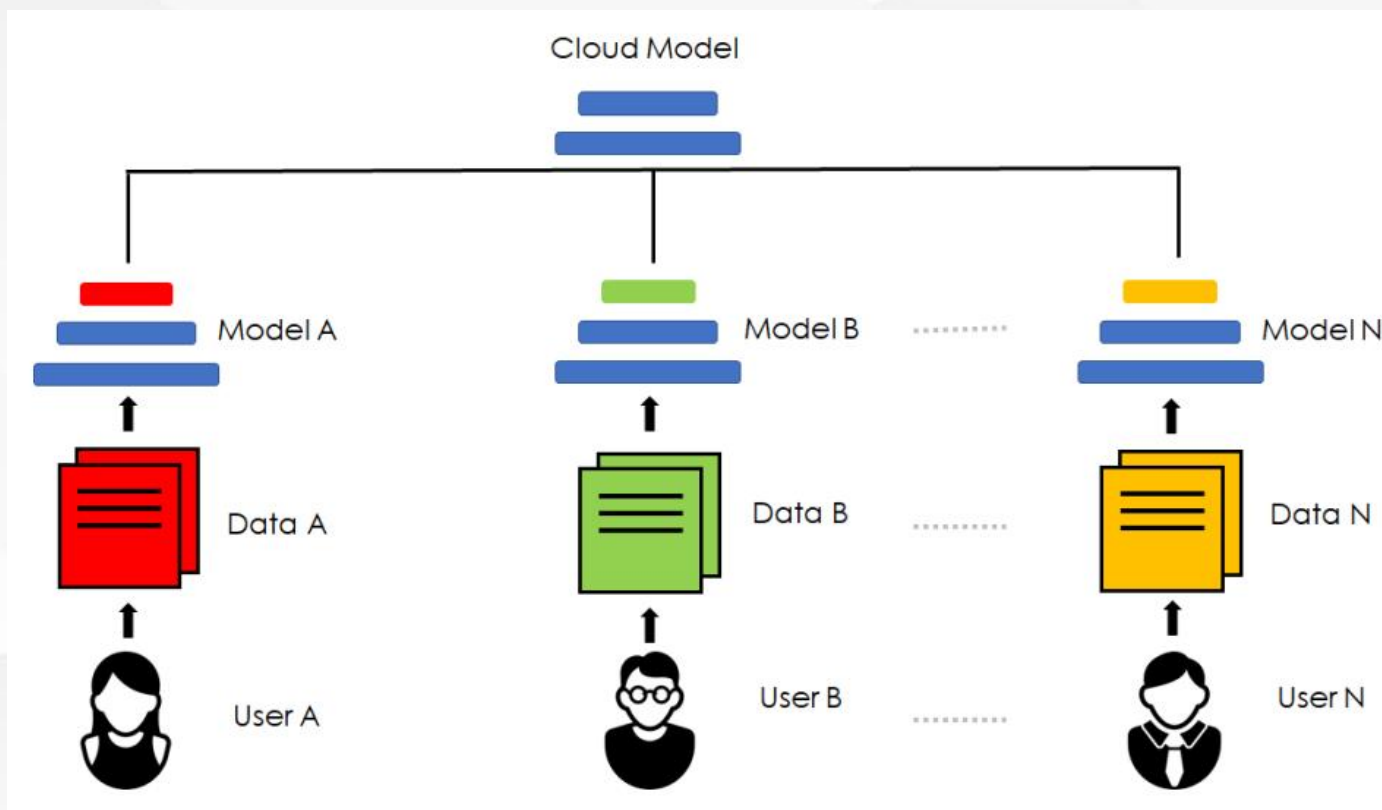




Heterogeneity

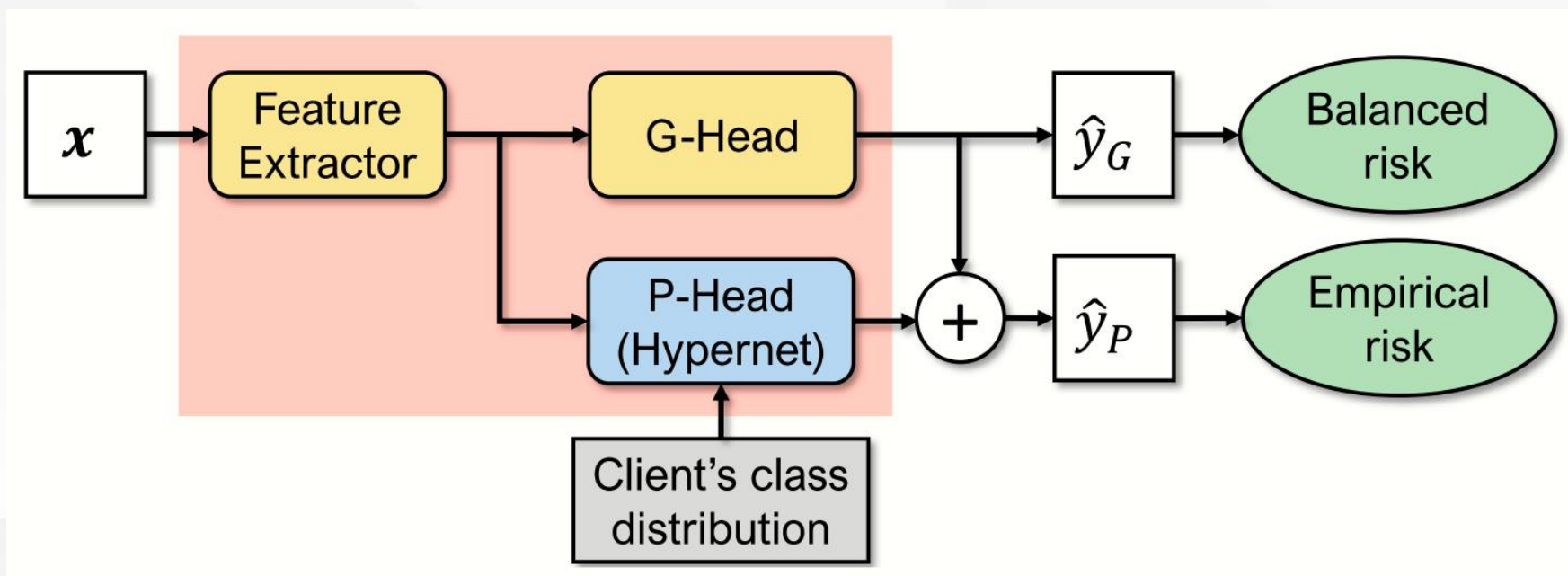
Personalized federated learning

- The base layers are shared with the parameter server while the personalization layers are kept private by each device.



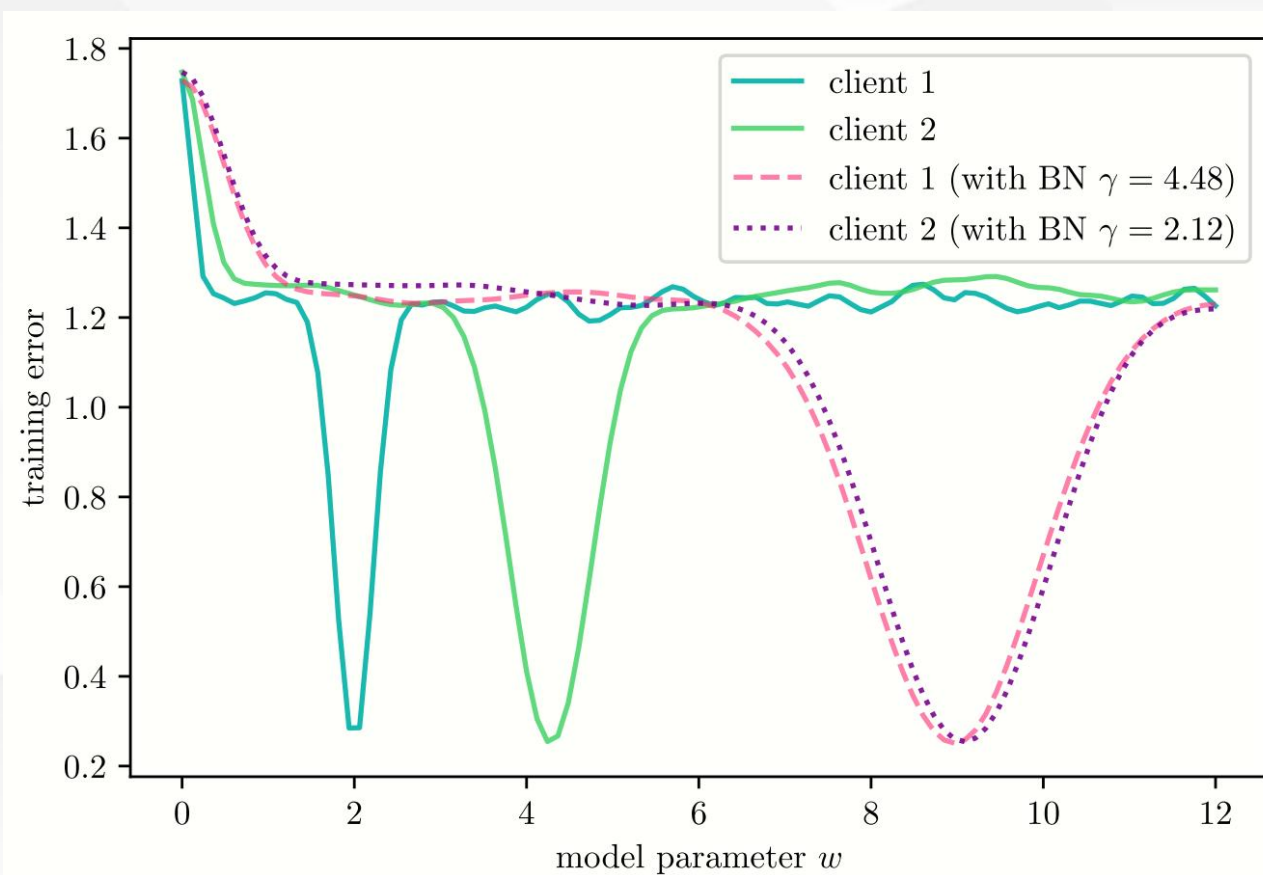
Personalized federated learning

- In addition to only learning for the local objective, FedRoD also proposes to simultaneously learn the balanced objective and the local objective on each client.



Personalized federated learning

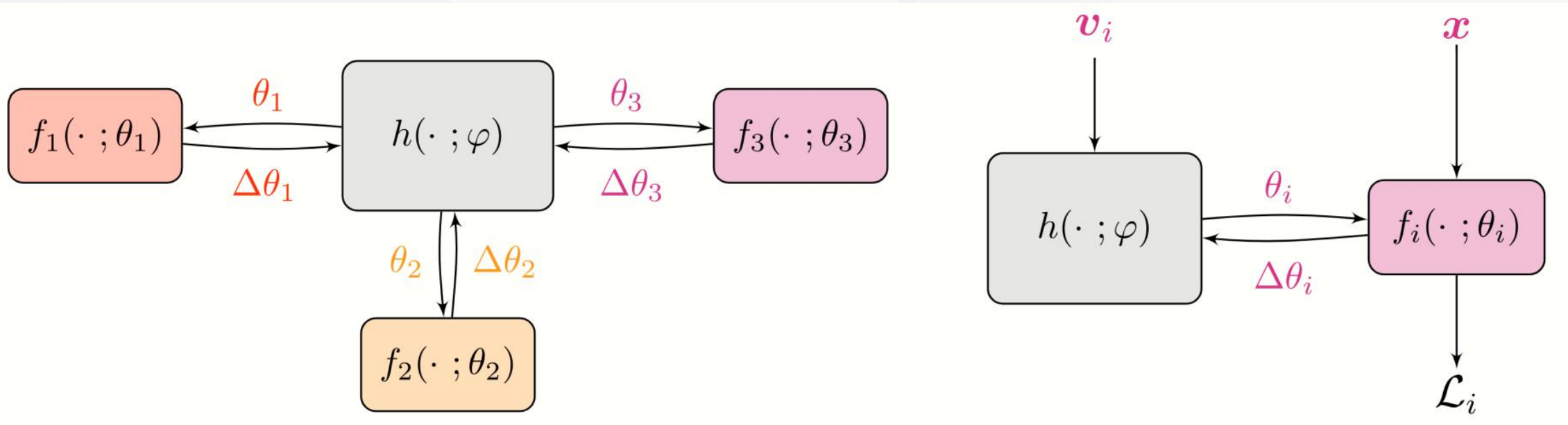
- Rather than localize only the higher layers, FedBN finds that the batch normalization (BN) layers in the ResNets are not beneficial for aggregation and proposes to localize all of them.





Personalized federated learning

- To further personalize models, FedHN assigns one client embedding for one client, and generate client model parameters through the hypernetwork on the server.

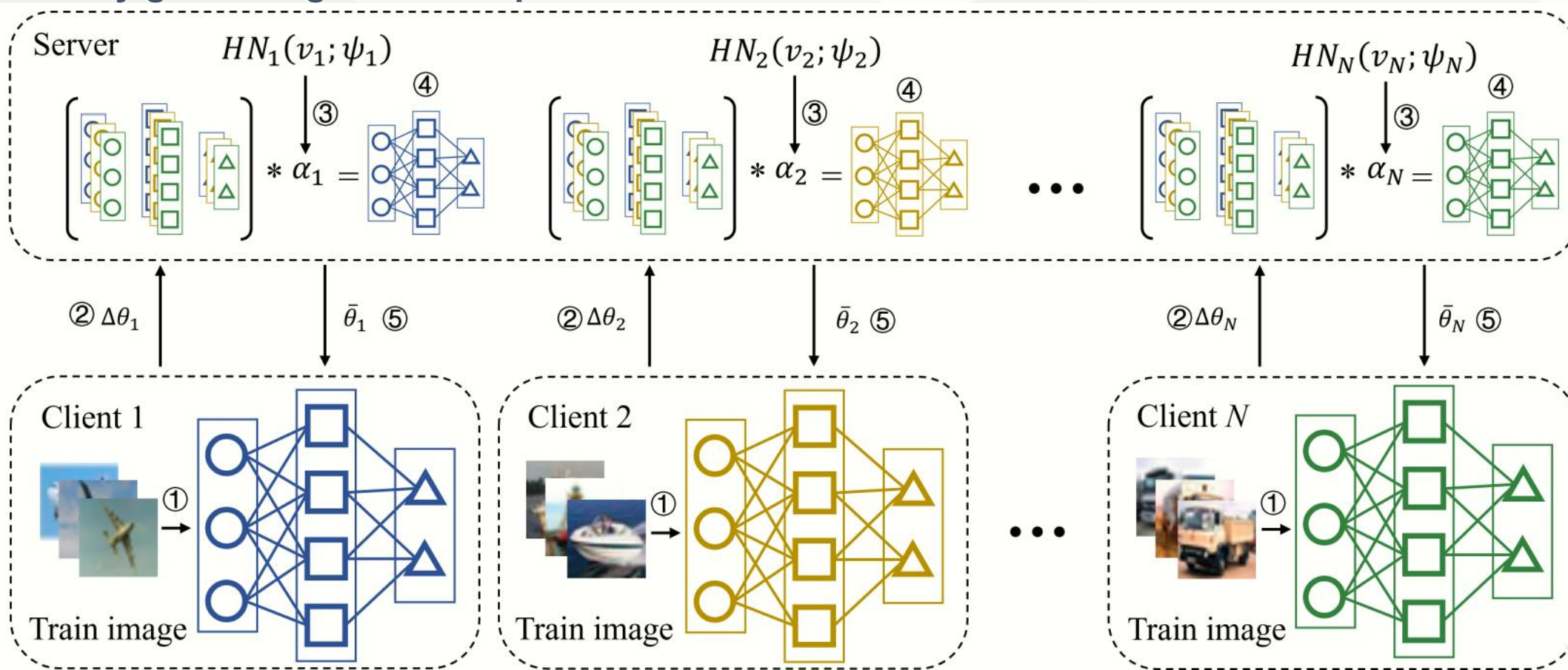


Updating the hypernetwork by client models

Generating client models

Personalized federated learning

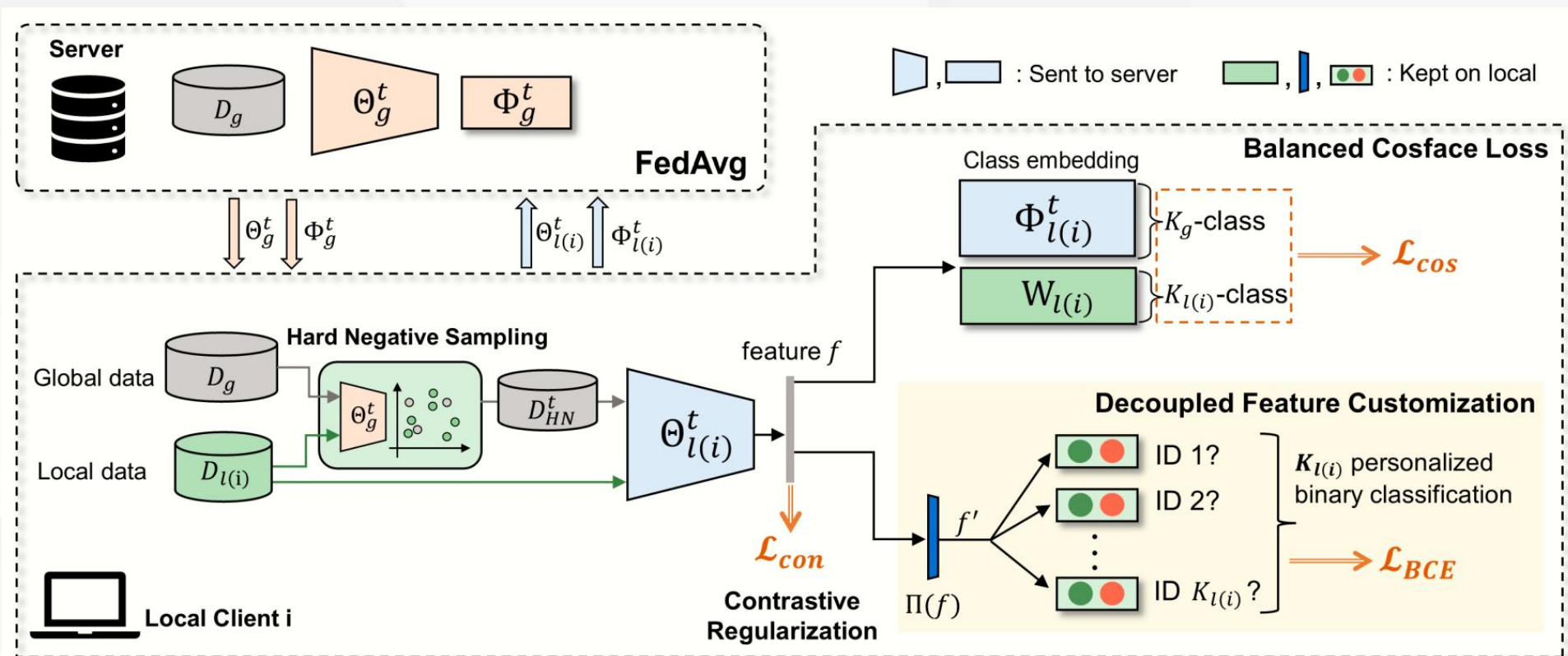
- With the client embeddings, pFedLA generate the layer-wise aggregation weights instead of directly generating the model parameters.



Heterogeneity

Personalized federated learning

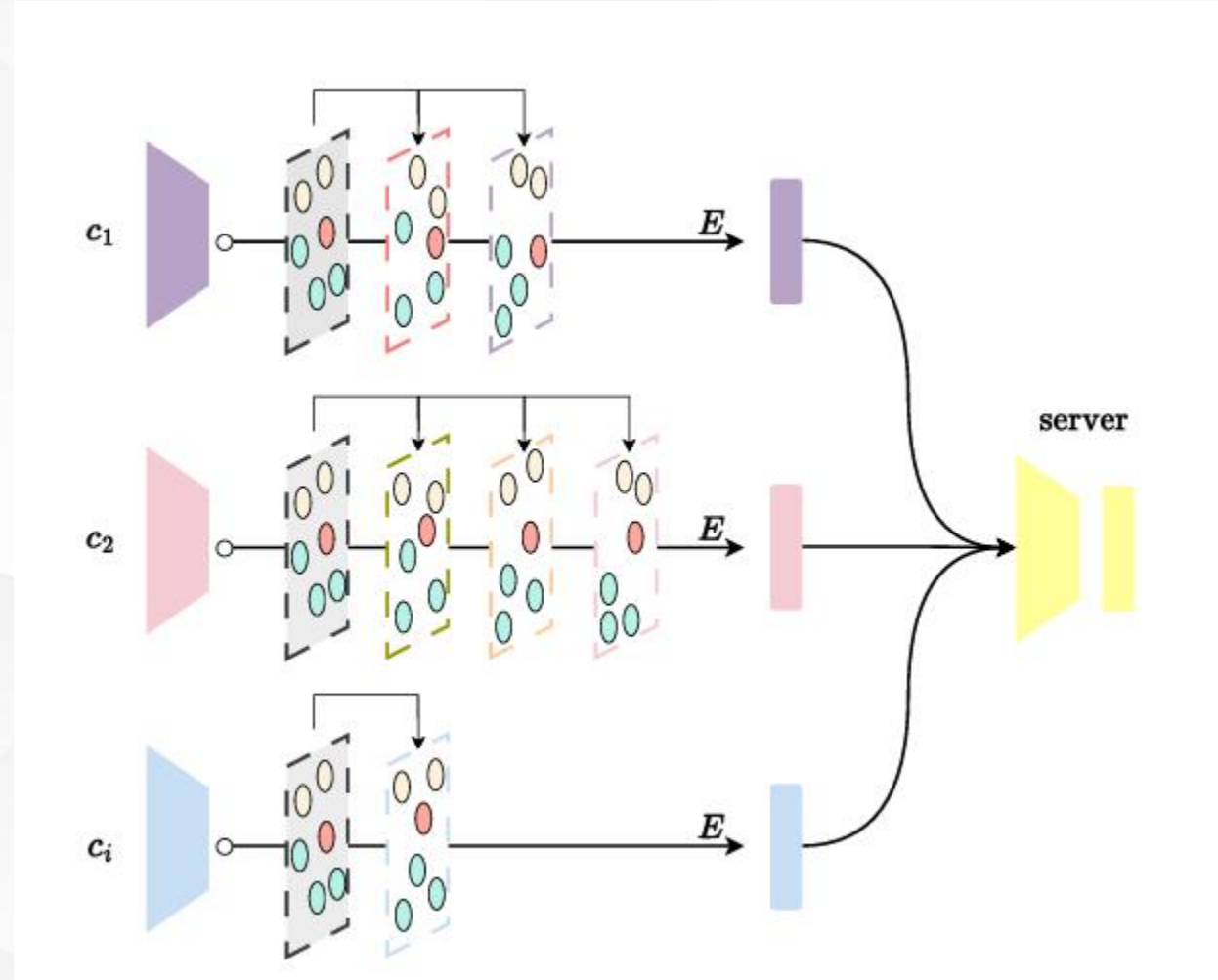
- Public dataset also helps FL
- Like FedRoD, FedFR learns two objectives on the client, one of which is the balanced objective, in the face recognition task.





Adaptive Local Epochs (ALE)

- Dynamically adjust local training epochs based on device capability:
 - High-end devices: 10-20 epochs
 - IoT devices: 1-2 epochs

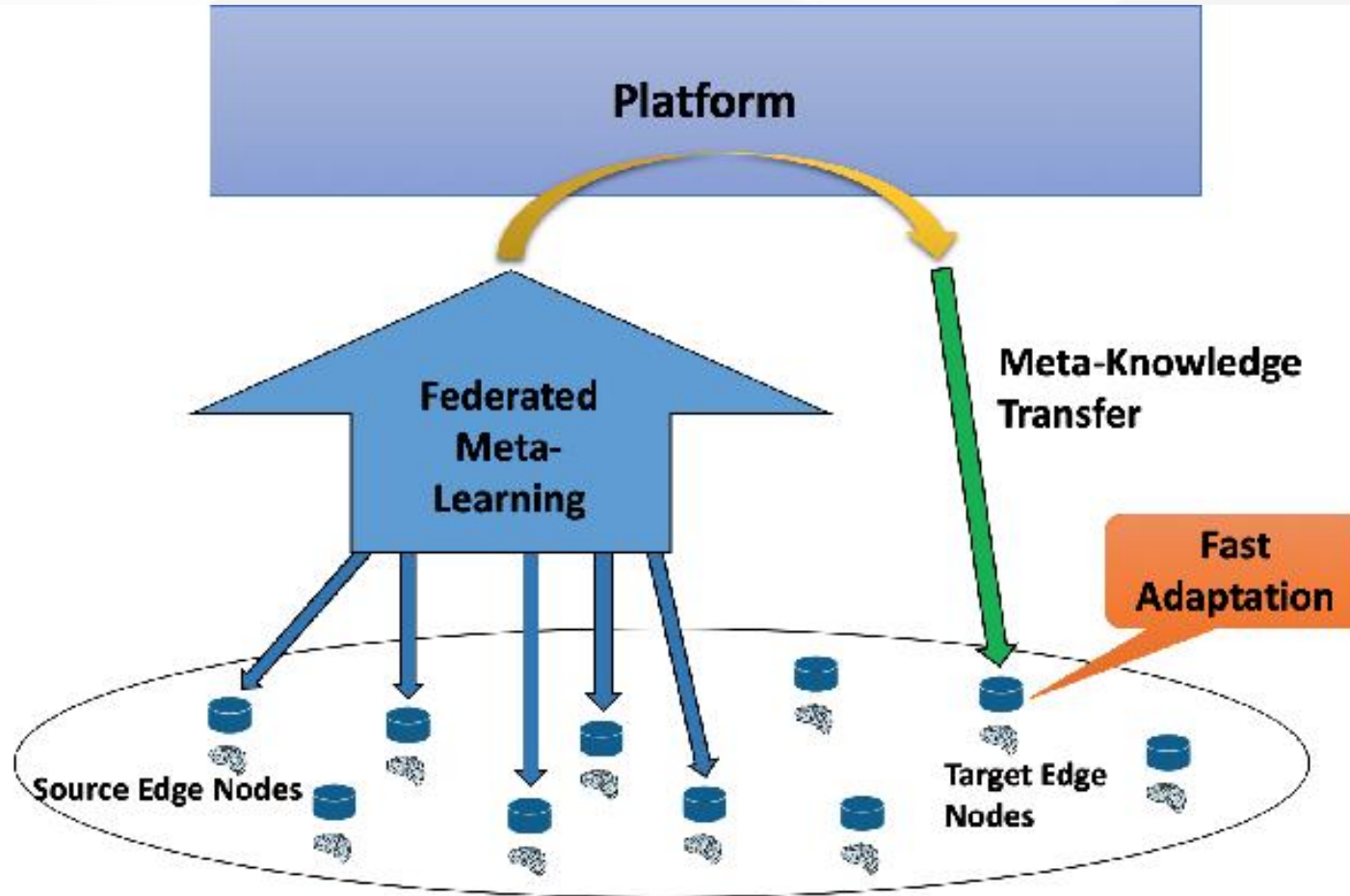




Advanced Heterogeneity Handling

⊗ Heterogeneous Federated Meta-Learning (HFML)

- Use meta-learning to initialize models for devices with non-IID data.



05

Application

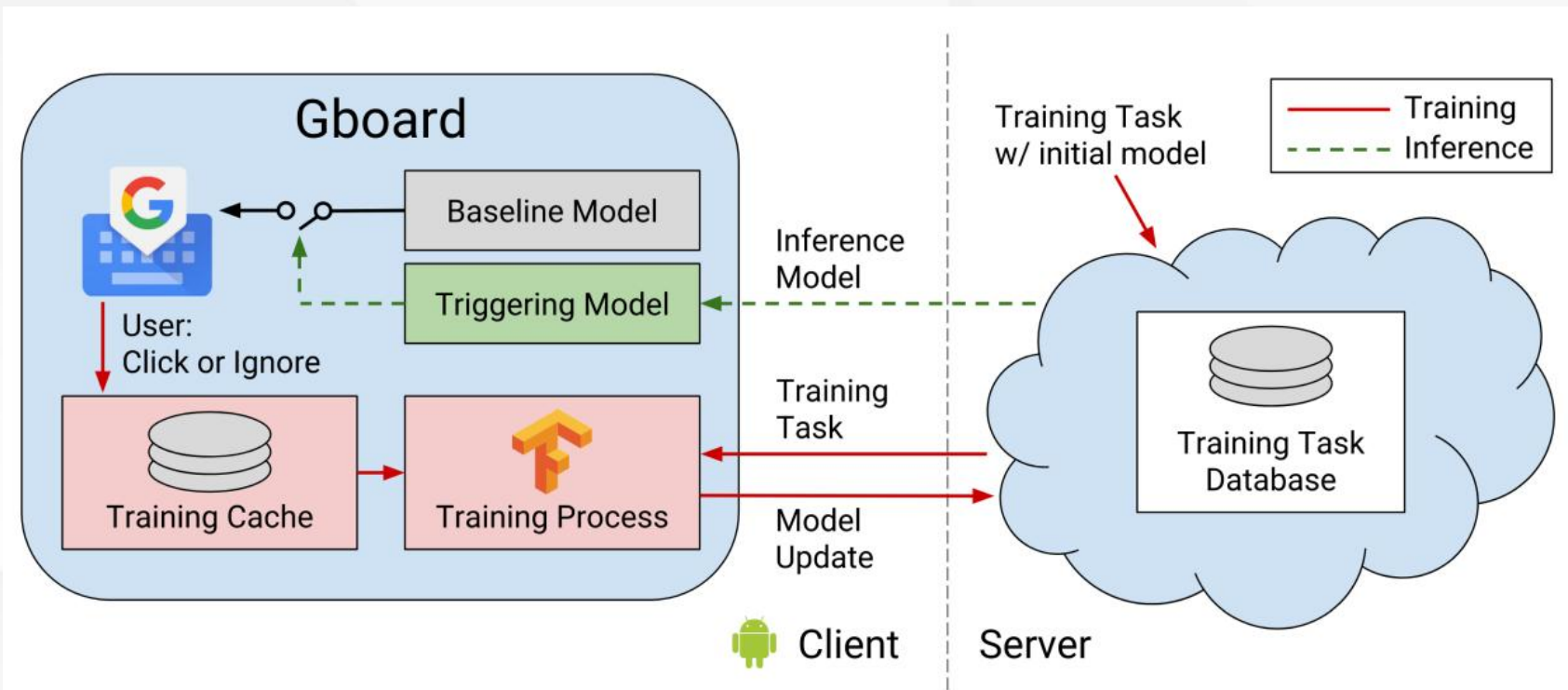
- Gboard
- Recommender system
- Autonomous driving
- ...



Application

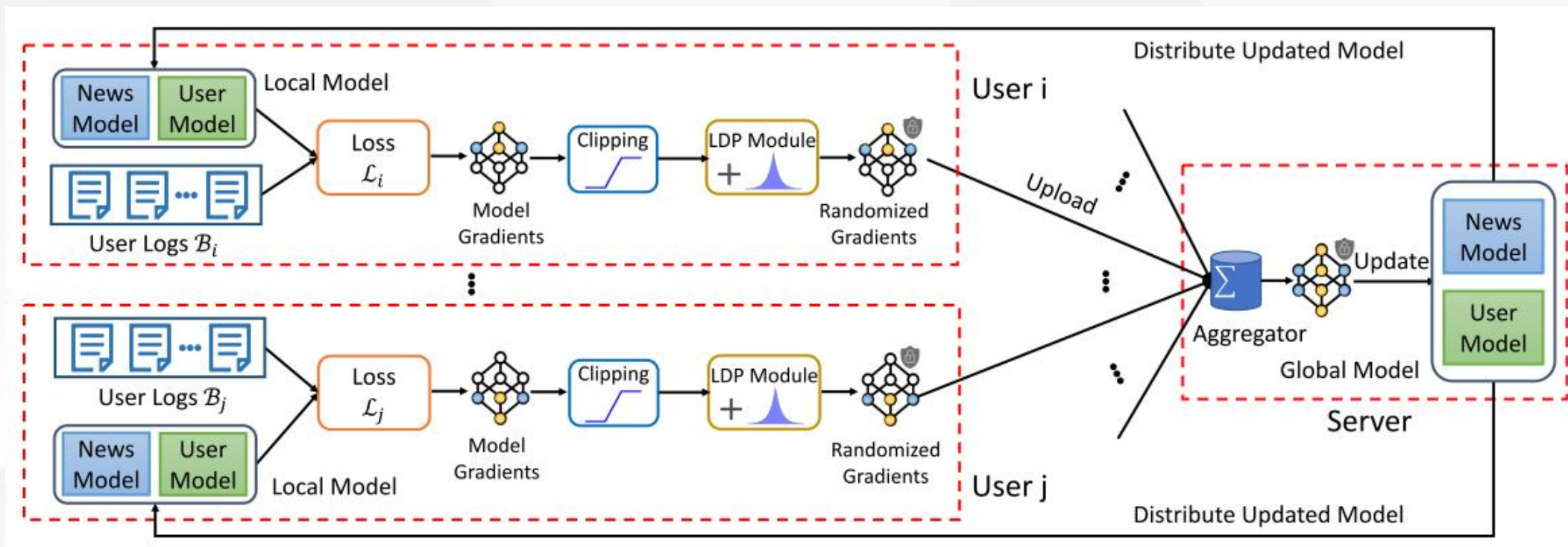
Gboard

- Google's first implementation of federated learning.
- Triggering model is trained federated to tune the results of the pre-trained baseline model for better performance.



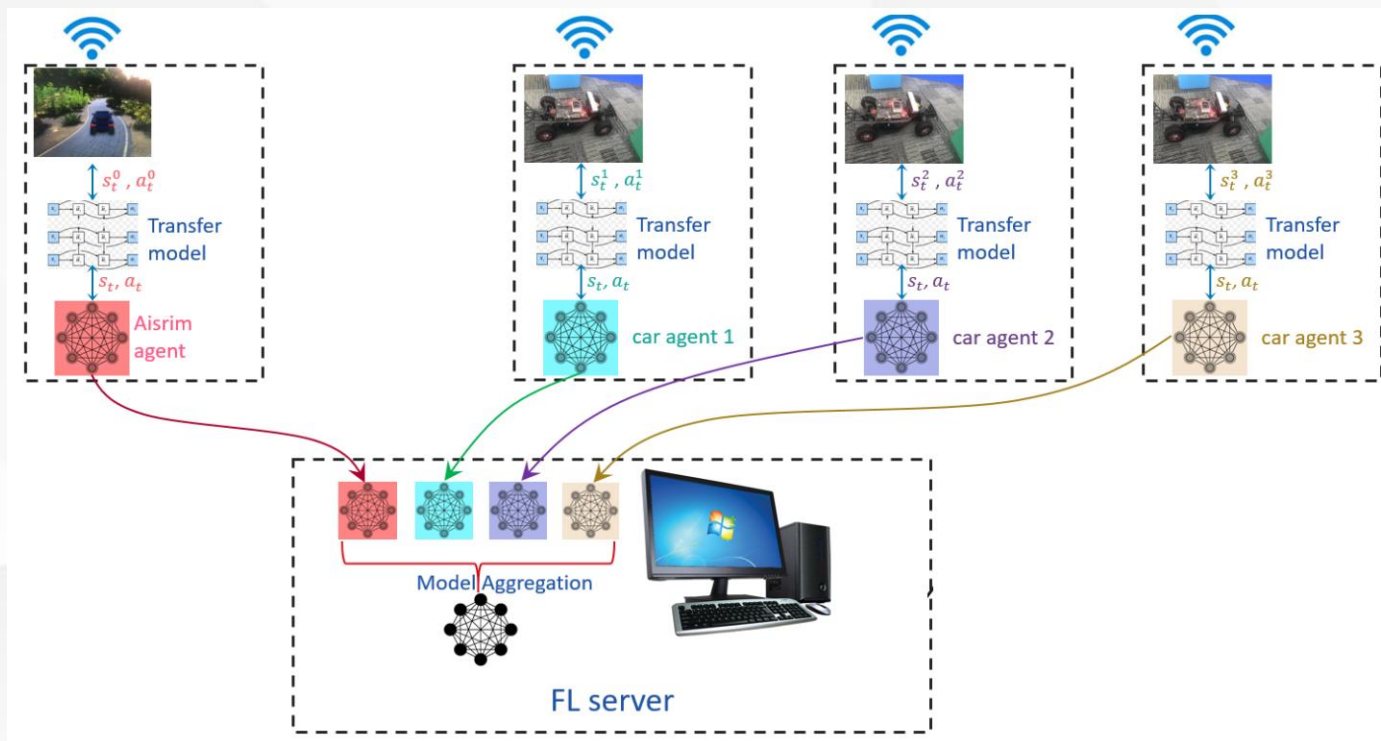
Recommender system

- The news model aims to learn news representations to model news content.
- The user model is used to learn user representations to model their personal interest.
- LDP denotes the local differential privacy



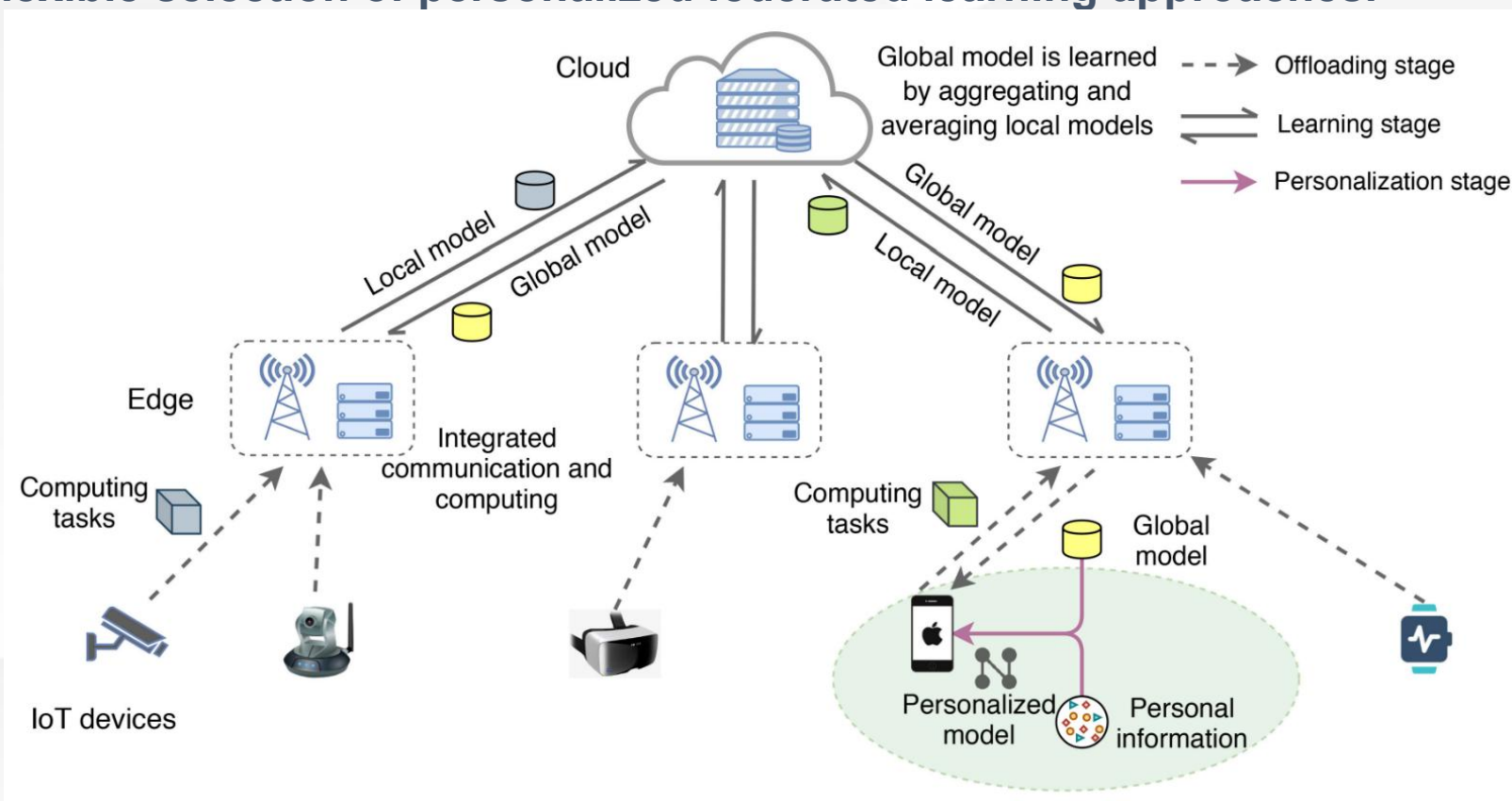
Autonomous driving

- The FTRL framework for collision avoidance RL tasks of autonomous driving cars
- Global model is asynchronously updated by different RL agents.
- Transfer knowledge from virtual world (Airsim platform) to real world



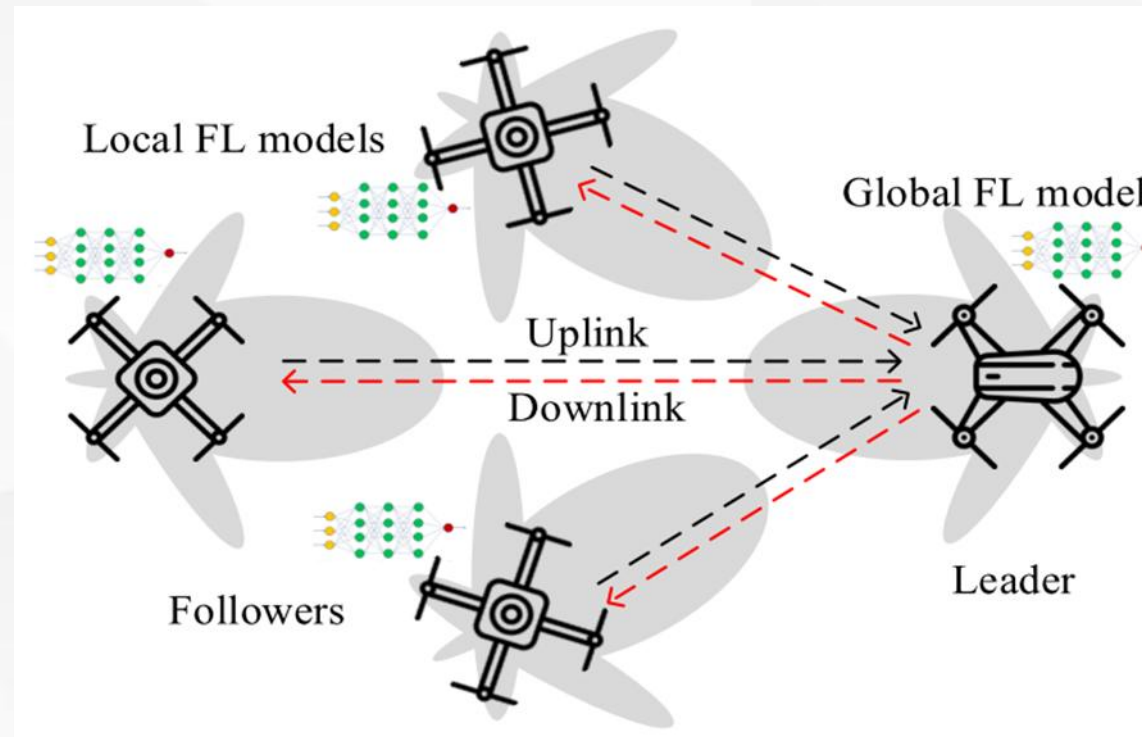
IOT

- Personalized federated learning framework for intelligent IoT applications.
- Supports flexible selection of personalized federated learning approaches.



UAV (Unmanned aerial vehicle)

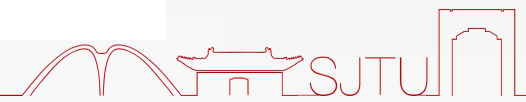
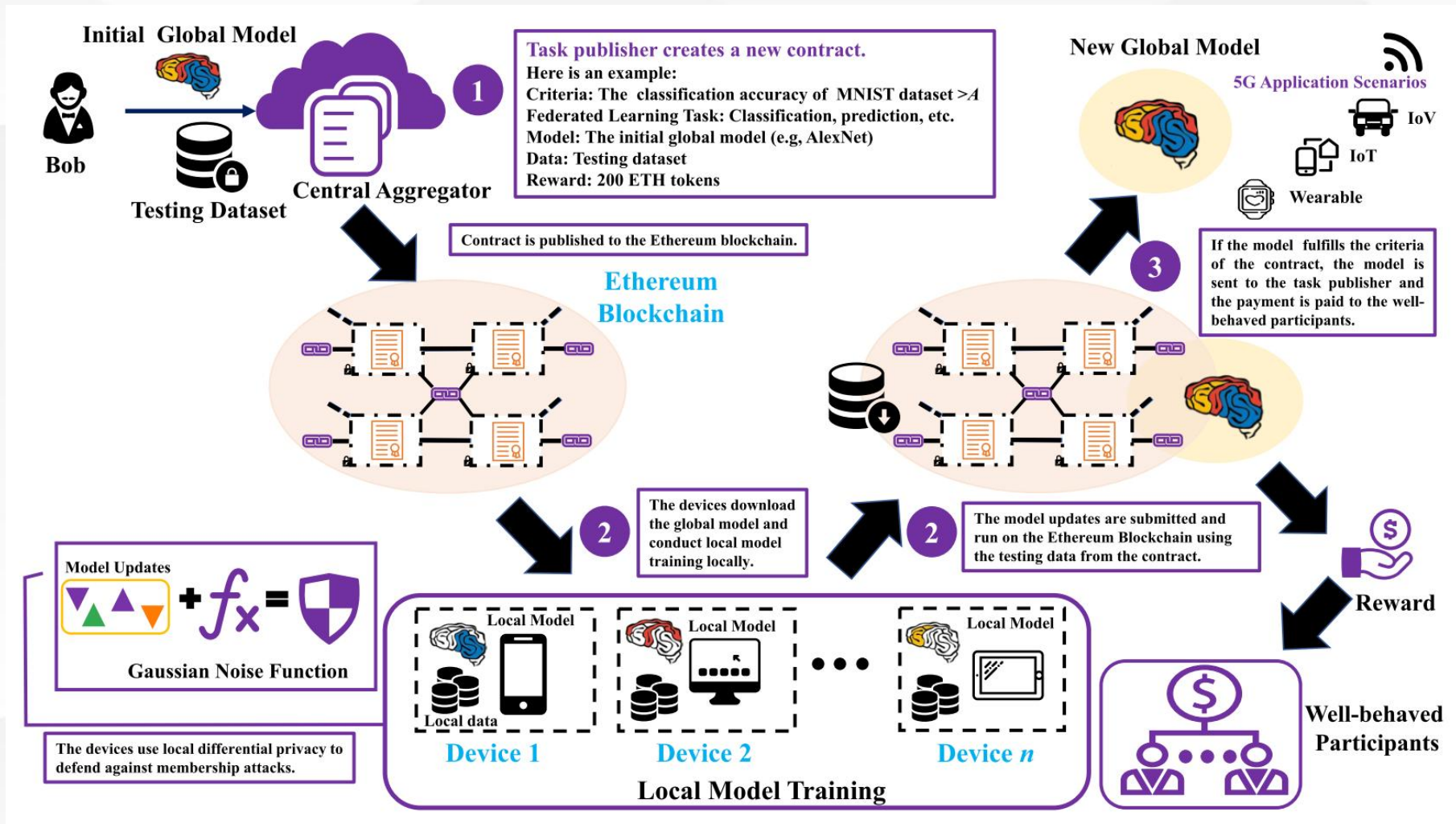
- Due to the high mobility of UAVs and their limited energy and stringent energy limitations, the analysis in previous federated learning work cannot be directly applied for UAV swarms.
- Use a sample average approximation approach from stochastic programming along with a dual method from convex optimization.





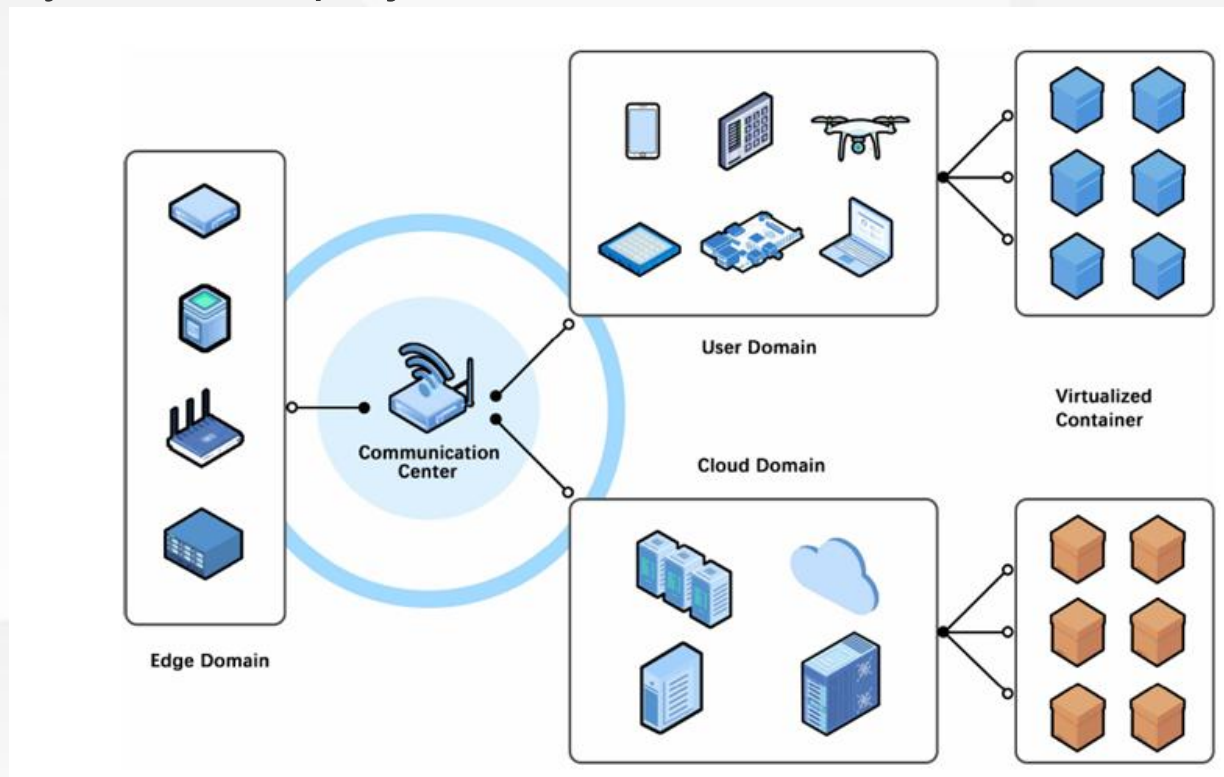
Application

Blockchain



Decentralized Finance (DeFi) Risk Modeling

- FL enables multiple banks to collaboratively train fraud detection models without sharing raw transaction data.
- Case Study: ChainFL project reduces DeFi hack detection time by 80%.





感谢聆听

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